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Inventor(s)	HARA; Kosuke et al.

WORKING MACHINE, INFORMATION PROCESSING DEVICE, AND PROGRAM

Abstract

An information processing device includes an operation planning part configured to determine a future operation of a working machine from among a plurality of operations according to a performance status of an operation of the working machine. A non-transitory computer-readable recording medium has a program embodied therein for causing an information processing device to determine a future operation of a working machine from among a plurality of operations according to a performance status of an operation of the working machine.

Inventors: **HARA; Kosuke (Kanagawa, JP), TSUZUKI; Ryuji (Kanagawa, JP)**

Applicant: **SUMITOMO HEAVY INDUSTRIES, LTD.** (Tokyo, JP)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of International Application No. PCT/JP2023/041864, filed on Nov. 21, 2023, and designated the U.S., which is based upon and claims priority to Japanese Patent Application no. 2022-186780, filed on Nov. 22, 2022. The entire contents of these applications are incorporated herein by reference.

BACKGROUND
[0002] The disclosures herein relate to working machines, and the like.
[0003] Working machines such as shovels are known in a related art.
SUMMARY
[0004] According to one embodiment of the present disclosure, an information processing device includes an operation planning part configured to determine a future operation of a working machine from among a plurality of operations according to a performance status of an operation of the working machine.
[0005] According to another embodiment of the present disclosure, a non-transitory computer-readable recording medium having a program embodied therein causes an information processing device to determine a future operation of a working machine from among a plurality of operations according to a performance status of an operation of the working machine.
[0006] According to further another embodiment of the present disclosure, a non-transitory computer-readable recording medium having a program embodied therein a support to perform causes device determining a future operation of a working machine from among a plurality of operations according to a performance status of an operation of the working machine, and notifying an operator of the determined future operation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0007] FIG. **1** is a drawing illustrating an example of an operation support system;
[0008] FIG. **2** is a top view illustrating an example of a shovel;
[0009] FIG. **3** is a drawing illustrating an example of a configuration related to remote control of the shovel;
[0010] FIG. **4** is a drawing illustrating an example of a hardware configuration of the shovel;
[0011] FIG. **5** is a drawing illustrating an example of a hardware configuration of an information processing device;
[0012] FIG. **6** is a functional block diagram illustrating a first example of a functional configuration of the operation support system;
[0013] FIG. **7** is a drawing describing an example of a relationship between a timing of processing in shovel operation planning and a planned operation;
[0014] FIG. **8** is a drawing describing another example of the relationship between the timing of processing in shovel operation planning and the planned operation;
[0015] FIG. **9** is a state transition diagram illustrating an example of a transition of a shovel operation in slope construction work;
[0016] FIG. **10** is a state transition diagram illustrating an example of a transition of a shovel operation in ground leveling work;

[0017] FIG. **11** is a functional block diagram illustrating a second example of the functional configuration of the operation support system;
[0018] FIG. **12** is a flowchart schematically illustrating an example of processing related to a start of an autonomous operation of the shovel;
[0019] FIG. **13** is a main flowchart schematically illustrating an example of processing related to shovel operation planning and bucket path generation;
[0020] FIG. **14** is a drawing illustrating an example of an observation target area;
[0021] FIG. **15** is a sub-flowchart schematically illustrating an example of processing related to the bucket path generation;
[0022] FIG. **16** is a drawing illustrating an example of cost conditions and operation parameters corresponding to a plurality of sections of a shovel excavation operation; and
[0023] FIG. **17** is a flowchart schematically illustrating an example of processing related to shovel operation control.

DETAILED DESCRIPTION

[0024] It is desirable for a working machine to perform a more appropriate operation according to various conditions from the viewpoint of work efficiency, for example.
[0025] Therefore, in view of the above problems, it is an object to provide a technology that enables the working machine to perform the more appropriate operation.
[0026] According to the embodiments of the present disclosure, the working machine can perform the more appropriate operation.
[0027] In the following, embodiments of the present disclosure will be described with reference to the accompanying drawings.

[Outline of Operation Support System]

[0028] An outline of an operation support system SYS according to the present embodiment will be described with reference to FIGS. **1** to **3**.
[0029] FIG. **1** is a drawing illustrating an example of the operation support system SYS. In FIG. **1**, a left side view of a shovel **100** is shown. FIG. **2** is a top view illustrating an example of the shovel **100**. FIG. **3** is a drawing illustrating an example of a configuration related to remote control of the shovel **100**. Hereinafter, a direction in the shovel **100**, or a direction viewed from the shovel **100** may be described by defining a direction in which an attachment AT extends in the top view of the shovel **100** (upward direction in FIG. **2**) as “front”.
[0030] As shown in FIG. **1**, the operation support system SYS includes the shovel **100**, an information processing device **200**, and a sensor group **300**.
[0031] The operation support system SYS supports the operation of the shovel **100** by coordinating with the shovel **100** using the information processing device **200**.
[0032] The shovel **100** included in the operation support system SYS may be one or a plurality of shovels.
[0033] The shovel **100** is a working machine to be supported in the operation support system SYS.
[0034] s shown in FIGS. **1** and **2**, the shovel **100** includes a lower traveling body **1**, an upper swivel body **3**, the attachment AT including a boom **4**, an arm **5**, and a bucket **6**, and a cabin **10**.
[0035] The lower traveling body **1** uses crawlers **1**C to cause the shovel **100** to travel. The crawlers **1**C include a left crawler **1**CL and a right crawler **1**CR. The crawler **1**CL is hydraulically driven by a traveling hydraulic motor **1**ML. Similarly, the crawler **1**CR is hydraulically driven by a traveling hydraulic motor **1**MR. Thus, the lower traveling body **1** can move by itself.
[0036] The upper swivel body **3** is mounted on the lower traveling body **1** via a swivel mechanism **2** so as to be capable of swiveling (freely swivel). For example, the upper swivel body **3** swivels with respect to the lower traveling body **1** by hydraulically driving the swivel mechanism **2** by a swivel hydraulic motor **2**M.
[0037] The boom **4** is attached to the front center of the upper swivel body **3** so that it is capable of lifting with respect to a rotation axis along the lateral direction. The arm **5** is rotatably attached to

the tip of the boom **4** around a rotation axis oriented laterally. The bucket **6** is rotatably attached to the tip of the arm **5** around a rotation axis oriented laterally.

[0038] The bucket **6** is an example of an end attachment, and is used, for example, for excavation work, slope work, and ground leveling work.

[0039] The bucket **6** is attached to the tip of the arm **5** in a manner that allows it to be easily replaced based on work requirements of the shovel **100**. That is, in place of the bucket **6**, a bucket of a different kind from the bucket **6**, for example, a relatively large bucket, a bucket for slope work, a bucket for dredging work, or the like may be attached to the tip of the arm **5**. An end attachment other than a bucket, for example, an agitator, a breaker, a crusher, or the like may be attached to the tip of the arm **5**. A spare attachment, for example, a quick coupling or a tilt rotator, may be provided between the arm **5** and the end attachment.

[0040] The boom **4**, the arm **5**, and the bucket **6** are hydraulically driven by the boom cylinder **7**, the arm cylinder **8**, and the bucket cylinder **9**, respectively.

[0041] The cabin **10** is a control cabin in which an operator rides and operates the shovel **100**. The cabin **10** is mounted, for example, on the front left of the upper swivel body **3**.

[0042] The shovel **100** includes as equipment a communication device **60** and can communicate with the information processing device **200** through a predetermined communication line NW.

[0043] The communication line NW includes, for example, a local area network (LAN) of a work site. The communication line NW may also include a wide area network (WAN). The wide area network includes, for example, a mobile communication network terminated at a base station, a satellite communication network using a communication satellite, and an Internet network. The communication line NW may also include, for example, a short-distance communication line based on a wireless communication standard such as WiFi or Bluetooth (registered trademark).

[0044] For example, the shovel **100** causes driven elements such as the lower traveling body **1** (i.e., a pair of right and left crawlers **1**CL and **1**CR), the upper swivel body **3**, the boom **4**, the arm **5**, and the bucket **6** to operate according to the operation of the operator in the cabin **10**.

[0045] In addition, the shovel **100** may be configured to be operable by the operator in the cabin **10**, or may be configured to be remotely operated (remote control) from the outside of the shovel **100**. When the shovel **100** is remotely operated, the interior of the cabin **10** may be unmanned. When the shovel **100** is dedicated to remote control, the cabin **10** may be omitted. Hereinafter, the description proceeds on the assumption that the operation of the operator includes at least one of the operation of an operating device **26** of the operator in the cabin **10** and the remote control of an external operator.

[0046] For example, as shown in FIG. **3**, the remote control includes a mode in which the shovel **100** is operated by the operation input related to an actuator of the shovel **100** performed by the remote control support device **400** which can communicate with the shovel **100** through the communication line NW. The remote control support device **400** may be provided separately from the information processing device **200** or may be the information processing device **200**.

[0047] The remote control support device **400** may be provided, for example, in a management center or the like which manages the work of the shovel **100** from the outside. Further, the remote control support device **400** may be a portable operation terminal, and in this case, the operator can perform the remote control of the shovel **100** while directly confirming the work status of the shovel **100** from surroundings of the shovel **100**.

[0048] The shovel may **100** transmit an image (hereinafter, “surrounding image”) representing the surrounding state including the front of the shovel **100** based on the captured image output by an imaging device mounted in the shovel to the remote control support device **400** through, for example, the communication device **60** described later. Further, the shovel **100** may transmit the captured image output from the imaging device to the remote control support device **400** through the communication device **60**, and the remote control support device **400** may process the captured image received from the shovel **100** to generate the surrounding image. Then, the remote control support device **400** may display the surrounding images representing the surrounding condition including the front of the shovel **100** on its own display device. In addition, various information images (information screens) displayed on an output device **50** (display device) inside the cabin **10** of the shovel **100** may similarly be displayed on the display device of the remote control support device **400**. Thus, the operator using the remote control support device **400** can remotely operate the shovel **100** while confirming, for example, the display contents of the information screen and the image representing the surrounding condition of the shovel **100** displayed on the display device. Then, the shovel **100** may operate the actuator to drive the driven elements such as the lower traveling body **1**, the upper swivel body **3**, the boom **4**, the arm **5**, and the bucket **6** according to the remote control signal representing the contents of the remote control received from the remote control support device **400** by the communication device **60**.

[0049] Further, the remote control may include a mode in which the shovel **100** is operated by, for example, external voice input or gesture input to the shovel **100** by a person (e.g., worker) around the shovel **100**. More specifically, the shovel **100** recognizes voices spoken by surrounding workers, gestures performed by workers, or the like, through a voice input device (e.g., microphone), a gesture input device (e.g., imaging device), or the like mounted in the shovel. Then, the shovel **100** may operate the actuator to drive the driven elements such as the lower traveling body **1** (right and left crawlers **1**C), the upper swivel body **3**, the boom **4**, the arm **5**, and the bucket **6** according to the contents of the recognized voices, gestures, or the like.

[0050] The shovel **100** may automatically operate the actuator regardless of the contents of the operator's operation. Thus, the shovel **100** can achieve a function of automatically operating at least a part of the driven elements such as the lower traveling body **1**, the upper swivel body **3**, and the attachment AT, that is, what is called an “automatic operation function” or a “Machine Control (MC) function”.

[0051] The automatic operation function includes, for example, a semi-automatic operation function (operation-assisted MC function). The semi-automatic operation function is a function of automatically operating driven elements (actuators) other than the driven element (actuator) to be operated in response to the operator's operation. The automatic operation function may also include a fully automatic operation function (fully automatic MC function). The fully automatic operation function is a function of automatically operating at least a part of the plurality of driven elements (hydraulic actuators) on the assumption that there is no operator's operation. When the fully automatic operation function is enabled in the shovel **100**, the cabin **10** may be in an unmanned state. When the shovel **100** is dedicated to fully automatic operation, the cabin **10** may be omitted. The semi-automatic operation function and the fully automatic operation function include, for example, a rule-based automatic operation function. The rule-based automatic operation function is an automatic operation function in which the operation contents of the driven element (actuator) to be operated automatically are determined according to a previously defined rule. The semi-automatic operation function and the fully automatic operation function may include an autonomous operation function. The autonomous operation function is an automatic operation function in which the shovel **100** autonomously makes various determinations, and the operation contents of the driven element (hydraulic actuator) to be operated automatically are determined according to the determination results.

[0052] In addition, the operation of the shovel **100** may be remotely monitored. In this case, a remote monitoring support device having the same function as the remote control support device **400** may be provided. The remote monitoring support device is, for example, an information processing device **200**. Thus, a monitor who is a user of the remote monitoring support device can monitor the operation of the shovel **100** while checking the surrounding image displayed on the display device of the remote monitoring support device. In addition, if the monitor determines that it is necessary from the viewpoint of safety, for example, the monitor can intervene in the operation or automatic operation of the shovel **100** by using the input device of the remote monitoring

support device and by making a predetermined input, and can make the shovel **100** stop urgently.
[0053] The information processing device **200** coordinates with the shovel **100** by communicating with each other and supports the operation of the shovel **100**.
[0054] The information processing device **200** is, for example, a server device or a terminal device for management installed in a management office in the work site of the shovel **100**, or in a management center or the like that manages the operation status of the shovel **100** at a place different from the work site of the shovel **100**. The server device may be an on-premises server, a cloud server, or an edge server. The terminal device for management may be, for example, a stationary terminal device such as a desktop PC (Personal Computer), or a portable terminal device (portable terminal) such as a tablet terminal, a smartphone, or a laptop PC. In the latter case, a worker at the work site, a supervisor who supervises the work, or a manager who manages the work site can carry the portable information processing device **200** and move around the work site. In the latter case, an operator can carry the portable information processing device **200** into the cabin of the shovel **100**, for example.
[0055] The information processing device **200** acquires data related to the operating state from the shovel **100**, for example. Thus, the information processing device **200** can determine the operating state of the shovel **100** and monitor presence or absence of abnormality of the shovel **100**. In addition, the information processing device **200** can display data related to the operating state of the shovel **100** through the display device **208** described later, for example, and allow the user to confirm the data. The information processing device **200** can, for example, make a learning model learn the operating state of the shovel **100** and generate a trained model for supporting the operation of the shovel **100**.
[0056] In addition, the information processing device **200** may transmit to the shovel **100** various data such as programs and reference data used in the processing of a controller **30** or the like to the shovel **100**. Thus, the shovel **100** can perform various processes related to the operation of the shovel **100** using various data downloaded from the information processing device **200**.
[0057] The sensor group **300** is installed at the work site of the shovel **100**. Target material includes, for example, earth in the work area around the shovel **100**.
[0058] For example, when a plurality of shovels **100** are included in the operation support system SYS, the sensor group **300** is provided for each shovel **100**. In addition, when a plurality of shovels **100** included in the operation support system SYS perform work at the same work site, one sensor group **300** may be shared for the plurality of shovels **100**.
[0059] The sensor group **300** includes sensors **300-1** to **300**-M (M: an integer of 2 or more). The sensors **300-1** to **300**-M measure the state of objects in the work site around the shovel **100** and acquire measurement data related to the state. The objects in the work site include, for example, target materials (earth in the work area) around the shovel **100**, working machines such as other shovels and bulldozers, and working vehicles such as trucks for transporting earth around the shovel **100**. The state of the objects includes a shape and characteristic of the objects.
[0060] The sensors **300-1** to **300**-M include, for example, ranging sensors (distance sensors). The ranging sensors include, for example, LIDAR (Light Detecting and Ranging), millimeter wave radar, ultrasonic sensors, infrared sensors, and the like. The sensors **300-1** to **300**-M may also include, for example, a stereo camera, a TOF (Time Of Flight) camera, and a 3D camera capable of acquiring data on distance (depth) in addition to two-dimensional images. The sensors **300-1** to **300**-M may also include both the ranging sensor and the 3D camera. Thus, the sensor group **300** can acquire measurement data representing the shape of the object at the work site around the shovel **100**. Hereinafter, a sensor capable of acquiring measurement data representing the shape of the object, such as a ranging sensor or a 3D camera, may be conveniently referred to as a “shape sensor”.
[0061] In addition, the sensors **300-1** to **300**-M may include a multi-wavelength spectral camera. The multi-wavelength spectral camera includes, for example, a multi-spectral camera or a

hyperspectral camera. Thus, for example, the sensor group **300** can acquire measurement data representing a characteristic of the object at the work site around the shovel **100**, such as hardness and moisture content of the earth. Hereinafter, a sensor capable of acquiring measurement data representing the characteristics of the object, such as a multi-wavelength spectral camera, may be conveniently referred to as a “characteristic sensor”.

[0062] For example, the sensors **300-1** to **300**-M include a plurality of shape sensors. The plurality of shape sensors may be provided at different locations in the work site around the shovel **100**, with their sensing ranges overlapping at least one other shape sensor's sensing range. Thus, for example, even if one shape sensor's measurement data cannot acquire data representing the shape of a part of an object within the sensing range due to occlusion, other shape sensors may be able to acquire the measurement data for the part of the object. Therefore, the sensor group **300** can more reliably acquire measurement data representing the shape of an object in the work site around the shovel **100**.

[0063] Further, the sensors **300-1** to **300**-M may include a plurality of characteristic sensors. Further, the plurality of characteristic sensors may be provided at different locations in the work site around the shovel **100**, with their sensing ranges overlapping at least one other characteristic sensor's sensing range. Thus, for example, even if one characteristic sensor's measurement data cannot acquire data representing the shape of a part of an object within the sensing range due to occlusion, other characteristic sensors may be able to acquire the measurement data for the part of the object. Therefore, the sensor group **300** can more reliably acquire measurement data representing the characteristics of an object in the work site around the shovel **100**.

[0064] In addition, the sensors **300-1** to **300**-M may include a sensor having both the function of a shape sensor and the function of a characteristic sensor. In this case, the sensors **300-1** to **300**-M may include a plurality of integrated sensors (hereinafter, “integrated sensor”). Further, the plurality of integrated sensors may be provided at different locations in the work site around the shovel **100**, with their sensing ranges overlapping at least one other integrated sensor's sensing range.

[0065] The sensor group **300** may simply include only one shape sensor or characteristic sensor. In place of the sensor group **300**, the operation support system SYS may simply include only one sensor capable of acquiring measurement data related to the state of an object in the work site around the shovel **100**.

[0066] The sensors **300-1** to **300**-M may be fixed to the work site around the shovel **100** or mounted on a mobile body movable in the work site around the shovel **100**. The mobile body may include, for example, a working machine or a working vehicle moving in the work site. The mobile body movable in the work site may include, for example, a flying body such as a drone flying over the work site.

[0067] The output (measured data) of the sensors **300-1** to **300**-M is received into the information processing device **200** through the communication line NW. The output of the sensors **300-1** to **300**-M is received directly into the information processing device **200** through the communication line NW, for example. The output of the sensors **300-1** to **300**-M may be received into the shovel **100** through the communication line NW and received into the information processing device **200** via the shovel **100**. When the sensors **300-1** to **300**-M are mounted on a predetermined device such as the above-described mobile device, the output of the sensors **300-1** to **300**-M may be received into the predetermined device and received into the information processing device **200** from the device.

[Hardware Configuration of Operation Support System]

[0068] Next, the hardware configuration of the operation support system SYS will be described with reference to FIGS. **4** and **5** in addition to FIGS. **1** to **3**.

[0069] The hardware configuration of the remote control support device **400** may be the same as that of the information processing device **200**. Therefore, illustration and description of the hardware configuration of the remote control support device **400** will be omitted.

<Hardware Configuration of Shovel>

[0070] FIG. **4** is a block diagram illustrating an example of a hardware configuration of a shovel **100**.

[0071] In FIG. **4**, a path through which mechanical power is transmitted is indicated by a double line, paths through which high-pressure hydraulic fluid for driving the hydraulic actuators flows are indicated by solid lines, paths through which pilot pressure is transmitted are indicated by broken lines, and paths through which electrical signals are transmitted are indicated by dotted lines.

[0072] The shovel **100** includes components such as a hydraulic drive system for hydraulically driving the driven element, an operation system for operating the driven element, user a interface system for exchanging information with the user, a communication system for communicating with the outside, and a control system for various controls.

«Hydraulic Drive System»

[0073] As shown in FIG. **4**, the hydraulic drive system of the shovel **100** includes hydraulic actuators HA for hydraulically driving each of the driven elements such as the lower traveling body **1** (left and right crawlers **1**C), the upper swivel body **3**, the boom **4**, the arm **5**, and the bucket **6**, as described above. The hydraulic drive system of the shovel **100** according to the present embodiment includes an engine **11**, a regulator **13**, a main pump **14**, and a control valve **17**.

[0074] The hydraulic actuators HA include such as traveling hydraulic motors **1**ML and **1**MR, a swivel hydraulic motor **2**M, a boom cylinder **7**, an arm cylinder **8**, and a bucket cylinder **9**.

[0075] In the shovel **100**, some or all of the hydraulic actuators HA may be replaced with an electric actuator. In other words, the shovel **100** may be a hybrid shovel or an electric shovel.

[0076] The engine **11** is a motor of the shovel **100** and a main power source in the hydraulic drive system. The engine **11** is, for example, a diesel engine using light oil as fuel. The engine **11** is mounted, for example, at the rear of the upper swivel body **3**. The engine **11** rotates at a predetermined target speed, for example, directly or indirectly controlled by the controller **30** to be described later, and drives the main pump **14** and a pilot pump **15**.

[0077] In place of or in addition to the engine **11**, another motor (e.g., electric motor) or the like may be mounted in the shovel **100**.

[0078] The regulator **13** controls (adjusts) a discharge amount of the main pump **14** under the control of the controller **30**. For example, the regulator **13** adjusts an angle (hereinafter, “tilt angle”) of a swash plate of the main pump **14** according to a control command from the controller **30**.

[0079] The main pump **14** supplies hydraulic fluid to the control valve **17** through a high-pressure hydraulic line. The main pump **14** is mounted, for example, at the rear of the upper swivel body **3** as well as the engine **11**. The main pump **14** is driven by the engine **11** as described above. The main pump **14** is, for example, a variable displacement hydraulic pump, and the stroke length of a piston is adjusted by adjusting the tilt angle of the swash plate by the regulator **13** under the control of the controller **30**, and the discharge flow rate and discharge pressure are controlled as described above.

[0080] The control valve **17** drives the hydraulic actuators HA in accordance with the operator's operation or remote control of the operating device **26** or an operation command corresponding to the automatic operation function. The control valve **17** is mounted, for example, in the center of the upper swivel body **3**. The control valve **17** is connected to the main pump **14** via a high-pressure hydraulic line as described above, and selectively supplies hydraulic oil supplied from the main pump **14** to the respective hydraulic actuators HA in accordance with the operator's operation or an operation command corresponding to the automatic operation function. Specifically, the control valve **17** includes a plurality of controlling valves (direction switching valves) for controlling the flow rate and direction of the hydraulic oil supplied from the main pump **14** to the hydraulic actuators HA.

«Operation System»

[0081] As shown in FIG. **4**, the operation system of the shovel **100** includes a pilot pump **15**, an operating device **26**, a hydraulic controlling valve **31**, a shuttle valve **32**, and a hydraulic controlling valve **33**.

[0082] The pilot pump **15** supplies the pilot pressure to various hydraulic devices via a pilot line **25**. The pilot pump **15** is mounted, for example, in the rear part of the upper swivel body **3**, similar to the engine **11**. The pilot pump **15** is, for example, a fixed displacement hydraulic pump, and is driven by the engine **11** as described above.

[0083] The pilot pump **15** may be omitted. In this case, after the relatively high pressure hydraulic fluid discharged from the main pump **14** is depressurized by a predetermined pressure reducing valve, the relatively low pressure hydraulic fluid may be supplied to various hydraulic devices as the pilot pressure.

[0084] The operating device **26** is provided near an operator's seat of the cabin **10**, and is used for an operator to operate various driven elements. Specifically, the operating device **26** is used for an operator to operate the hydraulic actuators HA that drives each driven element, and as a result, the operator can operate the driven element to be driven by the hydraulic actuators HA. The operating device **26** includes a pedal device and a lever device for operating each driven element (hydraulic actuator HA).

[0085] For example, as shown in FIG. **4**, the operating device **26** is a hydraulic pilot type. Specifically, the operating device **26** uses hydraulic fluid supplied from the pilot pump **15** through the pilot line **25** and a branched pilot line **25**A, and outputs a pilot pressure corresponding to the operation contents to a secondary pilot line **27**A. The pilot line **27**A is connected to one inlet port of the shuttle valve **32**, and is connected to the control valve **17** through a pilot line **27** connected to an outlet port of the shuttle valve **32**. Thus, the pilot pressure corresponding to the operation contents of various driven elements (hydraulic actuators HA) in the operating device **26** can be input to the control valve **17** through the shuttle valve **32**. Therefore, the control valve **17** can drive the hydraulic actuators HA according to the operation contents of the operating device **26** by the operator or the like.

[0086] The operating device **26** may be an electric type. In this case, the pilot line **27**A, the shuttle valve **32**, and the hydraulic controlling valve **33** are omitted. Specifically, the operating device **26** outputs an electric signal (hereinafter, “operation signal”) according to the operation contents, and the operation signal is received into the controller **30**. Then, the controller **30** outputs a control command according to the operation contents, that is, a control signal according to the operation contents for the operating device **26**, to the hydraulic controlling valve **31**. Thus, the pilot pressure corresponding to the operation contents of the operating device **26** are input from the hydraulic controlling valve **31** to the control valve **17**, and the control valve **17** can drive the respective hydraulic actuators HA according to the operation contents of the operating device **26**.

[0087] The controlling valve (direction switching valve) built in the control valve **17** and driving the respective hydraulic actuators HA may be an electromagnetic solenoid type. In this case, the operation signal output from the operating device **26** may be directly input to the control valve **17** (i.e., to the electromagnetic solenoid type controlling valve).

[0088] In addition, as described above, some or all of the hydraulic actuators HA may be replaced with electric actuators. In this case, the controller **30** may output a control command corresponding to the operation content of the operating device **26** or the remote control content defined by the remote control signal to the electric actuator or a driver for driving the electric actuator. When the shovel **100** is operated remotely, the operating device **26** may be omitted.

[0089] The hydraulic controlling valve **31** is provided for each driven element (hydraulic actuator HA) to be operated by the operating device **26** and for each driving direction (e.g., the upward and downward directions of the boom **4**) of the driven element (hydraulic actuator HA). For example, two hydraulic controlling valves **31** are provided for each double-acting hydraulic actuator HA for driving the lower traveling body **1**, the upper swivel body **3**, the boom **4**, the arm **5**, and the bucket **6**, and the like. The hydraulic controlling valve **31** may be provided, for example, in the pilot line

25B between the pilot pump **15** and the control valve **17**, and may be configured so that the flow path area (i.e., the cross-sectional area through which the hydraulic fluid can flow) thereof can be changed. Thus, the hydraulic controlling valve **31** can output a predetermined pilot pressure to a secondary pilot line **27**B by utilizing the hydraulic fluid of the pilot pump **15** supplied through the pilot line **25**B. Therefore, the hydraulic controlling valve **31** can indirectly apply a predetermined pilot pressure corresponding to a control signal from the controller **30** to the control valve **17** through the shuttle valve **32** between the pilot line **27**B and the pilot line **27**. Thus, for example, the controller **30** can cause the hydraulic controlling valve **31** to supply the pilot pressure corresponding to an operation command corresponding to the automatic operation function to the control valve **17** to achieve the operation of the shovel **100** by the automatic operation function.

[0090] Further, the controller **30** may control the hydraulic controlling valve **31** to achieve the remote control of the shovel **100**. Specifically, the controller **30** outputs to the hydraulic controlling valve **31** a control signal corresponding to the content of the remote control specified by the remote control signal received from the remote control support device **400** by the communication device **60**. Thus, the controller **30** can cause the hydraulic controlling valve **31** to supply the pilot pressure corresponding to the content of the remote control to the control valve **17** to achieve the operation of the shovel **100** based on the remote control of the operator.

[0091] When the operating device **26** is an electric type, the controller **30** directly supplies the pilot pressure corresponding to the operation content (operation signal) of the operating device **26** from the hydraulic controlling valve **31** to the control valve **17** to achieve the operation of the shovel **100** based on the operation of the operator.

[0092] The shuttle valve **32** has two inlet ports and one outlet port, and outputs hydraulic fluid having the higher pilot pressure among the pilot pressures input to the two inlet ports to the outlet port. Like the hydraulic controlling valve **31**, the shuttle valve **32** is provided for each driven element (hydraulic actuator HA) to be operated by the operating device **26** and for each driving direction of the driven element (hydraulic actuator HA). For example, two shuttle valves **32** are provided for each double-acting hydraulic actuator HA for driving the lower traveling body **1**, the upper swivel body **3**, the boom **4**, the arm **5**, and the bucket **6**. One of the two inlet ports of the shuttle valve **32** is connected to the secondary pilot line **27**A of the operating device **26** (specifically, the above-described lever device and pedal device included in the operating device **26**), and the other is connected to the secondary pilot line **27**B of the hydraulic controlling valve **31**. The outlet port of the shuttle valve **32** is connected to the pilot port of the corresponding control valve of the control valve **17** through the pilot line **27**. The corresponding control valve is a control valve for driving the hydraulic actuator HA to be operated by the above-described lever device or pedal device connected to one inlet port of the shuttle valve **32**. Therefore, these shuttle valves **32** can apply the higher of the pilot pressure of the secondary pilot line **27**A of the operating device **26** and the pilot pressure of the secondary pilot line **27**B of the hydraulic controlling valve **31** to the pilot port of the corresponding control valve. In other words, the controller **30** can control the corresponding control valve by outputting the pilot pressure higher than the pilot pressure of the secondary side of the operating device **26** from the hydraulic controlling valve **31** regardless of the operator's operation of the operating device **26**. Therefore, the controller **30** can control the operation of the driven elements (lower traveling body **1**, upper swivel body **3**, boom **4**, arm **5**, and bucket **6**) and achieve the automatic operation function and the remote control function regardless of the operation state of the operator with respect to the operating device **26**.

[0093] The hydraulic controlling valve **33** is provided in the pilot line **27**A connecting the operating device **26** and the shuttle valve **32**. The hydraulic controlling valve **33** is configured so that the flow path area can be changed, for example. The hydraulic controlling valve **33** operates according to a control signal input from the controller **30**. Thus, when the operating device **26** is operated by an operator, the controller **30** can forcibly reduce the pilot pressure output from the operating device **26**. Therefore, even when the operating device **26** is operated, the controller **30** can forcibly reduce or stop the operation of the hydraulic actuator HA corresponding to the operation of the operating device **26**. Moreover, even when the operating device **26** is operated, for example, the controller **30** can reduce the pilot pressure output from the operating device **26** to be lower than the pilot pressure output from the hydraulic controlling valve **31**. Therefore, by controlling the hydraulic controlling valve **31** and the hydraulic controlling valve **33**, the controller **30** can surely cause the desired pilot pressure to act on the pilot port of the control valve in the control valve **17** regardless of the operation content of the operating device **26**, for example. Therefore, by controlling the hydraulic controlling valve **33** in addition to the hydraulic controlling valve **31**, the controller **30** can more appropriately achieve the automatic operation function and the remote control function of the shovel **100**.

«User Interface System»

[0094] As shown in FIG. **4**, the user interface system of the shovel **100** includes an operating device **26**, an output device **50**, and an input device **52**.

[0095] The output device **50** outputs various types of information to a user of the shovel **100** (e.g., the operator in the cabin **10** or the external operator of the remote control) or to a person around the shovel **100** (e.g., a worker or a driver of a working vehicle). For example, the output device **50** includes a lighting device and a display device for outputting various types of information in a visual manner. The lighting device is, for example, a warning lamp (indicator lamp). The display device is, for example, a liquid crystal or display an organic EL (Electroluminescence) display. For example, as shown in FIG. **2**, the lighting device and the display device may be provided inside the cabin **10** and output various types of information in a visual manner to an operator or the like inside the cabin **10**. Further, the lighting apparatus and the display device may be provided, for example, on the lateral surface of the upper swivel body **3**, and may output various types of information in a visual manner to a worker or the like around the shovel **100**.

[0096] The output device **50** may also include a sound output device which outputs various types of information in an auditory method (see FIG. **7**). The sound output device may include, for example, a buzzer or a speaker. The sound output device may be provided, for example, inside or outside the cabin **10**, and may output various types of information in an auditory method to an operator inside the cabin **10** or to a person (a worker, etc.) around the shovel **100**.

[0097] The output device **50** may also include a device which outputs various types of information in a tactile manner such as vibration of the operator's seat.

[0098] The input device **52** receives various types of inputs from the user of the shovel **100**, and signals corresponding to the received inputs are received into the controller **30**. For example, as shown in FIG. **2**, the input device **52** is provided inside the cabin **10** and receives inputs from an operator inside the cabin **10**. The input device **52** may be provided, for example, on the lateral surface of the upper swivel body **3**, and may receive input from a worker or the like around the shovel **100**.

[0099] For example, the input device **52** may include an operation input device that accepts input by mechanical operation from a user. The operation input device may include a touch panel mounted on the display device, a touch pad installed around the display device, a button switch, a lever, a toggle, and a knob switch provided on the operating device **26** (lever device).

[0100] The input device **52** may also include a voice input device that accepts voice input from a user. The voice input device may include, for example, a microphone.

[0101] The input device **52** may also include a gesture input device that accepts gesture input from the user. The gesture input device may include, for example, an imaging device that captures a gesture performed by a user.

[0102] The input device **52** may also include a biometric input device that accepts biometric input of the user. The biometric input includes, for example, the input of biometric information as such the user's fingerprint or iris.

«Communication System»

[0103] As shown in FIG. **4**, the communication system of the shovel **100** according to the present embodiment includes a communication device **60**.
[0104] The communication device **60** is connected to an external communication line NW and communicates with a device provided separately from the shovel **100**. The device provided separately from the shovel **100** may include a device outside the shovel **100** and a portable terminal device (portable terminal) brought into the cabin **10** by the user of the shovel **100**. The communication device **60** may include, for example, a mobile communication module conforming to a standard such as 4G (Fourth Generation) or 5G (Fifth Generation). The communication device **60** may include, for example, a satellite communication module. The communication device **60** may include, for example, a WiFi communication module or a Bluetooth (registered trademark) communication module. The communication device **60** may include a plurality of communication devices according to the type of communication line NW when there are a plurality of communication lines NW that can be connected.
[0105] For example, the communication device **60** communicates with external devices such as the information processing device **200** and the remote control support device **400** in the work site through a local communication line constructed at the work site. The local communication line may be, for example, a mobile communication line using local 5G (what is called local 5G) constructed at the work site or a local network using WiFi6.
[0106] The communication device **60** may communicate with the information processing device **200** and the remote control support device **400** located outside the work site through a wide-area communication line including the work site, that is, a wide-area network.
«Control System»
[0107] As shown in FIG. **4**, the control system of the shovel **100** includes a controller **30**. The control system of the shovel **100** according to the present embodiment includes an operating pressure sensor **29**, a sensor **40**, and sensors S**1** to S**9**.
[0108] The controller **30** performs various controls related to the shovel **100**.
[0109] Functions of the controller **30** may be achieved by any hardware or any combination of hardware and software. For example, as shown in FIG. **4**, the controller **30** includes an auxiliary storage device **30**A, a memory device **30**B, a CPU (Central Processing Unit) **30**C, and an interface device **30**D connected by bus BS**1**.
[0110] The auxiliary storage device **30**A is a nonvolatile storage means and stores programs to be installed, and also stores necessary files, data, and the like. The auxiliary storage device **30**A may be, for example, an EEPROM (Electrically Erasable Programmable Read-Only Memory) or a flash memory.
[0111] The memory device **30**B loads the program of the auxiliary storage device **30**A so that the CPU **30**C can read it, for example, when a program start instruction is given. The memory device **30**B is, for example, an SRAM (Static Random Access Memory).
[0112] The CPU **30**C performs the program loaded into the memory device **30**B, for example, and implements various functions of the controller **30** according to the instruction of the program.
[0113] The interface device **30**D functions, for example, as a communication interface for connecting to the communication line inside the shovel **100**. The interface device **30**D may include a plurality of different types of communication interfaces according to the type of communication line to be connected.
[0114] The interface device **30**D also functions as an external interface for reading data from or writing data to the storage medium. The storage medium is, for example, a dedicated tool connected to a connector installed inside the cabin **10** by a detachable cable. The storage medium may be, for example, a general-purpose storage medium such as an SD memory card or a USB (Universal Serial Bus) memory. Thus, a program for realizing various functions of the controller **30** may be provided, for example, by a portable storage medium and installed in the auxiliary storage device **30**A of the controller **30**. The program may be downloaded from another computer (e.g.,

information processor **200**) outside the shovel **100** through the communication device **60** and installed in the auxiliary storage device **30**A.

[0115] It should be noted that a part of the functions of the controller **30** may be achieved by other controllers (control devices). In other words, the functions of the controller **30** may be achieved by a plurality of controllers installed in the shovel **100** in a distributed manner.

[0116] The operating pressure sensor **29** detects the pilot pressure on the secondary side (pilot line **27**A) of the hydraulic pilot type operating device **26**, that is, the pilot pressure corresponding to the operating state of each driven element (hydraulic actuator HA) in the operating device **26**. A detection signal of the pilot pressure corresponding to the operating state of each driven element (hydraulic actuator HA) in the operating device **26** by the operating pressure sensor **29** is received into the controller **30**.

[0117] When the operating device **26** is an electric type, the operating pressure sensor **29** is omitted. This is because the controller **30** can determine the operating state of each driven element through the operating device **26** based on the operating signal received from the operating device **26**.

[0118] The sensor **40** acquires, for example, measurement data related to the shape of an object around the shovel **100**.

[0119] For example, the sensor **40** is a shape sensor capable of acquiring measurement data representing the shape of an object around the shovel **100**, such as a distance measurement sensor or a 3D camera. In addition to the function of the shape sensor, the sensor **40** may be an integrated sensor having the function of a characteristic sensor capable of acquiring measurement data representing the characteristic of an object around the shovel **100**, such as a multi-wavelength spectral camera.

[0120] For example, as shown in FIG. **2**, the sensor **40** includes sensors **40**F, **40**B, **40**L, and **40**R. The sensor **40**F measures the state (shape and characteristics) of the object in front of the upper swivel body **3**. The sensor **40**B measures the state of the object in the upper swivel body **3**. The sensor **40**L measures the state of the object to the left of the upper swivel body **3**. The sensor **40**R measures the state of the object to the right of the upper swivel body **3**. Thus, in the top view of the shovel **100**, the sensor **40** can measure the state of the object in the whole circumference centering on the shovel **100**, that is, in the angular direction of 360 degrees. Hereinafter, the sensors **40**F, **40**B, **40**L, and **40**R may be collectively or individually referred to as “sensor **40**X”.

[0121] The output data (i.e., measurement data about the state of the object around the shovel **100**) of the sensor **40** (sensor **40**X) is received into the controller **30** through a one-to-one communication line or an in-vehicle network. Thus, for example, the controller **30** can determine the state of the shape and characteristics of the object around the shovel **100** based on the output data of the sensor **40**X.

[0122] Some or all of the sensors **40**B, **40**L and **40**R may be omitted.

[0123] The sensor S**1** is attached to the boom **4** and measures an attitude state of the boom **4**. The sensor S**1** outputs measurement data indicating the attitude state of the boom **4**. The attitude state of the boom **4** is, for example, the attitude angle (hereinafter, “boom angle”) around the rotation axis of the base end corresponding to a connecting part of the boom **4** with the upper swivel body **3**. The sensor S**1** includes, for example, a rotary potentiometer, a rotary encoder, an acceleration sensor, an angular acceleration sensor, a six-axis sensor, an IMU (Inertial Measurement Unit), and the like. Hereinafter, the same may apply to the sensors S**2** to S**4**. The sensor S**1** may also include a cylinder sensor for detecting the expansion or contraction position of the boom cylinder **7**. Hereinafter, the same may apply to the sensors S**2** and S**3**. The output of the sensor S**1** (measurement data representing the attitude state of the boom **4**) is received into the controller **30**. Thus, the controller **30** can determine the attitude state of the boom **4**.

[0124] The sensor S**2** is attached to the arm **5** and measures the attitude state of the arm **5**. The sensor S**2** outputs measurement data indicating the attitude state of the arm **5**. The attitude state of the arm **5** is, for example, the attitude angle (hereinafter, "arm angle") around the rotational axis of the base end corresponding to the connecting part of the arm **5** with the boom **4**. The output of the sensor S**2** (measurement data indicating the attitude state of the arm **5**) is received into the controller **30**. Thus, the controller **30** can determine the attitude state of the arm **5**.

[0125] The sensor S**3** is attached to the bucket **6** and measures the attitude state of the bucket **6**. The sensor S**3** outputs measurement data indicating the attitude state of the bucket **6**. The attitude state of the bucket **6** is, for example, the attitude angle (hereinafter, "arm angle") around the rotation axis of the base end corresponding to the connecting part with the arm **5** of the bucket **6**. The output of the sensor S**3** (measurement data indicating the attitude state of the bucket **6**) is received into the controller **30**. Thus, the controller **30** can determine the attitude state of the bucket **6**.

[0126] The sensor S**4** measures the attitude state of the body (e.g., upper swivel body **3**) of the shovel **100**. The sensor S**4** outputs measurement data indicating the attitude state of the body of the shovel **100**. The attitude state of the body of the shovel **100** is, for example, the inclination state of the body with respect to a predetermined reference plane (e.g., horizontal plane). For example, the sensor S**4** is attached to the upper swivel body **3**, and measures the inclination angle of the shovel **100** about two axes in the longitudinal direction and the lateral direction (hereinafter, "longitudinal inclination angle" and "lateral inclination angle"). The output of the sensor S**4** (measurement data indicating the attitude state of the body of the shovel **100**) is received into the controller **30**. Thus, the controller **30** can determine the attitude state (inclination state) of the body (upper swivel body **3**).

[0127] The sensor S**5** is attached to the upper swivel body **3** and measures the swivel state of the upper swivel body **3**. The sensor S**5** outputs measurement data representing the swivel state of the upper swivel body **3**. The sensor S**5** measures, for example, a swivel angular velocity and a swivel angle of the upper swivel body **3**. The sensor S**5** includes, for example, a gyro sensor, a resolver, a rotary encoder, etc. The output of sensor the S**5** (measurement data representing the swivel state of the upper swivel body **3**) is received into the controller **30**. Thus, the controller **30** can determine the swivel state such as the swivel angle of the upper swivel body **3**.

[0128] The controller **30** can determine (estimate) the position of the tip (bucket **6**) of the attachment AT based on the outputs of the sensors S**1** to S**5**.

[0129] When the sensor S**4** includes a gyro sensor capable of detecting three-axis angular velocity, a six-axis sensor, an IMU, or the like, the swivel state (e.g., swivel angular velocity) of the upper swivel body **3** may be detected based on the detection signal of the sensor S**4**. In this case, the sensor S**5** may be omitted.

[0130] The sensor S**6** measures the position of the shovel **100**. The sensor S**6** may measure the position in world (global) coordinates or in local coordinates at the work site. In the former case, the sensor S**6** may be, for example, a GNSS (Global Navigation Satellite System) sensor. In the latter case, the sensor S**6** is a transceiver capable of communicating with a device that serves as a reference for the position at the work site and outputting a signal corresponding to the position relative to the reference. The output of the sensor S**6** is received into the controller **30**.

[0131] The sensor S**7** measures pressure (cylinder pressure) in an oil chamber of the boom cylinder **7**. The sensor S**7** includes, for example, a sensor for measuring the cylinder pressure (rod pressure) of the oil chamber on the rod side of the boom cylinder **7** and a sensor for measuring the cylinder pressure (bottom pressure) of the oil chamber on the bottom. The output of the sensor S**7** (measurement data of the cylinder pressure of the boom cylinder **7**) is received into the controller **30**.

[0132] The sensor S**8** measures the pressure (cylinder pressure) of the oil chamber of the arm cylinder **8**. The sensor S**8** includes, for example, a sensor for measuring the cylinder pressure (rod pressure) of the oil chamber on the rod side of the arm cylinder **8** and a sensor for measuring the cylinder pressure (bottom pressure) of the oil chamber on the bottom side of the arm cylinder **8**.

The output of the sensor S**8** (measurement data of the cylinder pressure of the arm cylinder **8**) is received into the controller **30**.
[0133] The sensor S**9** measures the pressure (cylinder pressure) of the oil chamber of the bucket cylinder **9**. The sensor S**9** includes, for example, a sensor for measuring the cylinder pressure (rod pressure) of the oil chamber on the rod side of the bucket cylinder **9** and a sensor for measuring the cylinder pressure (bottom pressure) of the oil chamber on the bottom side of the bucket cylinder **9**. The output of the sensor S**9** (measurement data of the cylinder pressure of the bucket cylinder **9**) is received into the controller **30**.
[0134] The controller **30** can determine the load state acting on the attachment AT based on the outputs of the sensors S**7** to S**9**. The load acting on the attachment AT includes, for example, the reaction force acting on the bucket **6** from the target material (earth) and the weight of the earth contained in the bucket **6**.
<Hardware Configuration of Information Processing Device>
[0135] FIG. **5** is a drawing illustrating an example of a hardware configuration of the information processing device **200**.
[0136] The functions of the information processing device **200** are achieved by optional hardware or combinations of optional hardware and software. For example, as shown in FIG. **5**, the information processing device **200** includes an external interface **201**, an auxiliary storage device **202**, a memory device **203**, a CPU **204**, a high speed arithmetic unit **205**, a communication interface **206**, an input device **207**, a display device **208**, and a sound output device **209**. These devices are connected by bus BS**2**.
[0137] The external interface **201** functions as an interface for reading data from a storage medium **201**A and writing data to the storage medium **201**A. The storage medium **201**A includes, for example, a flexible disk, CD (Compact Disc), DVD (Digital Versatile Disc), BD (Blu-ray (registered trademark) Disc), SD memory card, USB memory, and the like. Thus, the information processing device **200** can read various data to be used in processing through the storage medium **201**A, store them in the auxiliary storage device **202**, and install programs to achieve various functions.
[0138] The information processing device **200** may acquire various data and programs to be used in processing from an external device through the communication interface **206**.
[0139] The auxiliary storage device **202** stores various installed programs, and also stores files, data, and the like necessary for various processing. The auxiliary storage device **202** includes, for example, hard disc drives (HDDs), solid state discs (SSDs), flash memories, and the like.
[0140] The memory device **203** reads and stores a program from the auxiliary storage device **202** when a program start instruction is given. The memory device **203** includes, for example, DRAM (Dynamic Random Access Memory) and SRAM.
[0141] The CPU **204** performs various programs loaded from the auxiliary storage device **202** to the memory device **203**, and implements various functions related to the information processing device **200** according to the programs.
[0142] The high speed arithmetic unit **205** performs arithmetic processing at a relatively high speed in conjunction with the CPU **204**. The high speed arithmetic unit **205** includes, for example, GPU (Graphics Processing Unit), ASIC (Application Specific Integrated Circuit), and FPGA (Field-Programmable Gate Array).
[0143] Note that, the high speed arithmetic unit **205** may be omitted depending on the required arithmetic processing speed.
[0144] The communication interface **206** is used as an interface to connect to enable to communicate with an external device. Thus, the information processing device **200** can communicate with an external device such as a shovel **100**, for example, through the communication interface **206**. The communication interface **206** may have a plurality of types of communication interfaces depending on a communication system or the like with the connected

device.
[0145] The input device **207** receives various inputs from the user. The input device **207** includes an operating device for remote control of the shovel **100**.
[0146] The input device **207** includes, for example, an input device (hereinafter, "operation input device") which accepts mechanical operation input from the user. The operating device for remote control may be an operation input device. The operation input device includes, for example, a button, a toggle, a lever, a keyboard, a mouse, a touch panel mounted on the display device **208**, a touch pad provided separately from the display device **208**, and the like.
[0147] The input device **207** may also include a voice input device capable of receiving voice input from a user. The voice input device may include, for example, a microphone capable of collecting user voice.
[0148] The input device **207** may also include a gesture input device capable of receiving gesture input from a user. The gesture input device may include, for example, a camera capable of capturing an image of a user's gesture.
[0149] The input device **207** may also include a biometric input device capable of receiving biometric input from a user. The biometric input device may include, for example, a camera capable of acquiring image data containing information about a user's fingerprint or iris.
[0150] The display device **208** displays an information screen and an operation screen to the user of the information processing device **200**. The display device **208** is, for example, a liquid crystal display or an organic EL (Electroluminescence) display.
[0151] The sound output device **209** transmits various types of information by sound to the user of the information processing device **200**. The sound output device **209** is, for example, a buzzer, alarm, speaker, etc.
[Functional Configuration of Operation Support System]
[0152] Next, a functional configuration of the operation support system SYS will be described with reference to FIGS. **6** and **7** in addition to FIGS. **1** to **5**.
First Example
[0153] FIG. **6** is a functional block diagram illustrating a first example of the functional configuration of the operation support system SYS.
[0154] Hereinafter, a term "path of the work part of the shovel **100**" includes both a route where the work part of the shovel **100** has already moved (i.e., path) and a route where it may move in the future. The work part corresponds to the tip of the attachment AT used for applying a change with respect to the target material. Specifically, the work part is the bucket **6**.
[0155] The shovel **100** includes a support device **150**. In this example, the support device **150** supports the shovel **100** operated by the autonomous operation function in performing the work.
[0156] As shown in FIG. **6**, in this example, the support device **150** includes a controller **30**, a hydraulic controlling valve **31**, a sensor **40**, an output device **50**, an input device **52**, and sensors S**1** to S**9**. Further, when instructions concerning autonomous operation are input from outside the shovel **100**, the support device **150** may include a communication device **60** instead of or in addition to the input device **52**.
[0157] The controller **30** includes, as functional parts, an operation log providing part **301** and a work support part **302**.
[0158] When there are multiple shovels **100** included in the operation support system SYS, there may be a shovel **100** in which the controller **30** includes only the operation log providing part **301** and the work support part **302**, and a shovel **100** in which only the latter is included. In this case, the former shovel **100** has only the function of acquiring an operation log of the shovel **100** used for the work support function of the latter shovel **100** and providing it to the information processing device **200**. The same may be applied to the case of the second example described later.
[0159] The information processing device **200** includes, as functional parts, a log acquisition part **2001**, a simulator part **2002**, a log storage part **2003**, a training data generation part **2004**, a

machine learning part **2005**, a trained model storage part **2006**, and a distribution part **2007**.
[0160] The operation log providing part **301** is a functional part for acquiring the operation log of a predetermined operation of the shovel **100** and providing it to the information processing device **200**.
[0161] For each type of work, a plurality of (types of) predetermined operations are previously defined. For example, in the case of excavation work, the plurality of predetermined operations include an excavation operation, a boom raising and swivel operation, a boom lowering and swivel operation, an earth removing operation, a broom operation, and the like. In the case of ground leveling work, the plurality of predetermined operations include an excavation operation, an earth removing operation, a sweeping operation, a horizontal pulling operation, a rolling operation, a broom operation, and the like. In the case of slope work, the plurality of predetermined operations may include an excavation operation, an earth removing operation, a slope pulling operation, a rolling operation, and the like. The slope pulling operation corresponds to the horizontal pulling operation of the ground leveling work, and involves moving the attachment AT to draw a cutting edge (edge of teeth) of the bucket **6** toward the body (upper swivel body **3**) along the slope corresponding to the target construction surface. The sweeping operation involves, for example, operating the attachment AT and pushing the bucket **6** forward along the ground to sweep the earth forward at the back of the bucket **6**.
[0162] In the sweeping operation, for example, the attachment AT performs a lowering operation of the boom **4** and an opening operation of the arm **5**. The horizontal pulling operation involves, for example, operating the attachment AT and moving the edge of teeth of the bucket **6** along the ground drawing toward the front approximately horizontally to level unevenness of the ground (surface of terrain). In the horizontal pulling operation, for example, the attachment AT performs a raising operation of the boom **4** and a closing operation of the arm **5**. The rolling operation involves, for example, operating the attachment AT and pressing the ground against the back of the bucket **6**. Moreover, the rolling operation may involve pressing the ground by striking the back of the bucket **6** against the ground while moving the bucket **6** up and down. Further, the rolling operation may involve pushing the ground against the back of the bucket **6** forward along the ground so that the earth is swept out to a predetermined position forward by the back of the bucket **6**, and then pressing the ground against the ground with the back of the bucket **6**. In the rolling operation, for example, the attachment AT performs a lowering operation of the boom **4** when pressing the ground. The broom operation involves, for example, operating the upper swivel body **3** to swivel the bucket **6** left and right while keeping it along the ground. The broom operation may involve operating the attachment AT and the upper swivel body **3** to swivel the bucket **6** left and right while keeping it along the ground and push forward the bucket **6**. In the broom operation, for example, the upper swivel body **3** alternately repeats left and right swivel operations. In the broom operation, for example, in addition to the alternating left and right swivel operation of the upper swivel body **3**, the attachment AT may perform the lowering operation of the boom **4** and the opening operation of the arm **5**.
[0163] The operation log of the shovel **100** is time-series data representing the operation state of the shovel **100**. For example, the operation log of the shovel **100** includes time-series data representing the operation contents of the operator. The time-series data representing the operation contents of the operator includes, for example, time-series output data of the operating pressure sensor **29** corresponding to the hydraulic pilot type operating device **26** and time-series output data (operation signal data) of the operating device **26** corresponding to the electric type operating device **26**. The operation log of the shovel **100** may be time-series output data of the sensors S**1** to S**5** and time-series data representing the attitude state of the shovel **100** obtained from the output data of the sensors S**1** to S**5**.
[0164] In addition, the operation log providing part **301** may acquire the operation log of the shovel **100** operated by an operator (hereinafter, “skilled person” for convenience) having a long operating

history of the shovel **100** and provide the operation log to the information processing device **200**. Thus, the trained model LM**3** capable of reproducing the operation of the shovel **100** operated by an expert can be generated by machine learning based on the operation log of the shovel **100**, as described later.
[0165] The operation log providing part **301** includes an operation log recording part **301**A, an operation log storage part **301**B, and an operation log transmission part **301**C.
[0166] An operation log recording part **301**A acquires an operation log during a predetermined operation of the shovel **100** and records it in an operation log storage part **301**B. For example, each time a predetermined operation of the shovel **100** is performed, the operation log recording part **301**A records an operation log during the operation in an operation log storage part **301**B.
[0167] The operation log storage part **301**B stores the operation log of the shovel **100**. For example, for each predetermined operation performed by the shovel **100**, the operation log and data of the time (date and time) when the predetermined operation was performed are associated and stored in the operation log storage part **301**B. The data of the time when the predetermined operation was performed includes data of both the start and the end of the predetermined operation of the shovel **100**. When a plurality of predetermined operations are defined, for each predetermined operation performed by the shovel **100**, the operation log, data of the time when the predetermined operation was performed, and data of identification information of the performed predetermined operation are associated and stored in the operation log storage part **301**B. Hereinafter, data linked to the operation log of the shovel **100** may be conveniently referred to as “accompanying data”. For example, for each predetermined operation performed by the shovel **100**, record data representing a correspondence between the operation log and the accompanying is data accumulated in the operation log storage part **301**B, so that a database of the operation log at the time of performance of the predetermined operation of the shovel **100** is constructed.
[0168] The operation log of the operation log storage part **301**B which has already been transmitted to the information processing device **200** by the operation log transmission part **301**C described later may be deleted afterwards.
[0169] The operation log transmission part **301**C transmits the operation log stored in the operation log storage part **301**B when the shovel **100** performs a predetermined operation and the accompanying data associated with the operation log to the information processing device **200** through the communication device **60**. The operation log transmission part **301**C may also transmit to the information processing device **200** record data showing the correspondence between the operation log of the shovel **100** and the accompanying data for each predetermined operation performed by the shovel **100**.
[0170] For example, the operation log transmission part **301**C transmits a non-transmitted operation log of the shovel **100** and the accompanying data stored in the operation log storage part **301**B to the information processing device **200** in response to a transmission request of the operation log of the shovel **100** received from the information processing device **200**. The operation log transmission part **301**C may also automatically transmit the non-transmitted operation log of the shovel **100** and the accompanying data stored in the operation log storage part **301**B to the information processing device **200** at a predetermined timing. The predetermined timing is, for example, when the shovel **100** stops operation (key switch is turned off) or starts operation (key switch is turned on).
[0171] The log acquisition part **2001** acquires a log when the shovel **100** performs a predetermined operation.
[0172] The log when the shovel **100** performs the predetermined operation includes the operation log when the shovel **100** performs the predetermined operation and the state log of the target material. The state log of the target material includes time-series data showing the state of the target material before, during, and after the performance of the predetermined operation of the shovel **100**. The state of the target material includes a shape of the earth (terrain shape) and a characteristic

of the earth of the target material, and the like. The characteristics of the earth may include, for example, hardness of the earth, moisture content of the earth, size of the earth grain (grain size), angle of repose of the earth, and the like. The operation log when the shovel **100** performs the predetermined operation is uploaded from the shovel **100**. The state log of the target material when the shovel **100** performs the predetermined operation is acquired based on the measurement data uploaded from the sensor group **300** and the accompanying data (data at the time when the predetermined operation is performed) uploaded from the shovel **100**.

[0173] The state log of the target material may be acquired based on the measurement data of the sensor **40** of the shovel **100**. In this case, the measurement data acquired by the sensor **40** at the time of the predetermined operation of the shovel **100** is uploaded from the shovel **100** to the information processing device **200**. In this case, the sensor group **300** may be omitted.

[0174] The simulator part **2002** performs computer simulation of a predetermined operation of the shovel **100** using a virtual model of the shovel **100** and the target material (earth).

[0175] For example, by using a Distinct Element Method (DEM), the earth of the target material is modeled as a collection of minute particles. Thus, the simulator part **2002** causes the virtual model of the shovel **100** to perform a predetermined operation such as an excavation operation, and can virtually reproduce the whole behavior of the earth of the target material and the reaction forces from the earth in entirety by analyzing the individual operation of the minute particles.

[0176] The simulator part **2002** acquires data on the path of the work part of the shovel **100** and data on the state of the target material (earth) before, during, and after the performance of the predetermined operation as a log when the shovel **100** performs the predetermined operation by computer simulation. The former data corresponds to the operation log when the shovel **100** performs the predetermined operation by computer simulation, and the latter data corresponds to the state log of the target material when the shovel **100** performs the predetermined operation by computer simulation.

[0177] The simulator part **2002** performs computer simulation of a large number of patterns concerning a predetermined operation of the shovel **100** using various conditions of target materials (earth) and paths of various work parts of the shovel **100**. Thus, the simulator part **2002** can store logs when the shovel **100** performs a predetermined operation by computer simulation under different conditions in the log storage part **2003**.

[0178] Logs when the shovel **100** performs a predetermined operation acquired by the log acquisition part **2001** and the simulator part **2002** are stored in the log storage part **2003** in an accumulated form. For example, the log storage part **2003** stores an operation log for each predetermined operation actually performed by the shovel **100** or by computer simulation, a state log of the target material, and accompanying data in a form linked to each other. In the log storage part **2003**, the logs acquired by the log acquisition part **2001** and the logs acquired by the simulator part **2002** may be stored in a distinguishable manner or may be stored in a mixed manner in an indistinguishable manner.

[0179] The training data generation part **2004** generates training data for machine learning based on the logs when the shovel **100** performs a predetermined operation stored in the log storage part **2003**, and outputs a training data set which is an aggregate of a large number of training data. The training data generation part **2004** may automatically generate the training data by batch processing, or may generate the training data in response to an input from the user of the information processing device **200**. The training data generation part **2004** includes training data generation parts **2004**A to **2004**C.

[0180] The training data generation part **2004**A generates a training data set for generating the trained model LM**1**. The trained model LM**1** infers a future state of the target material of the shovel **100** at a predetermined time in the future by using a current state of the target material of the shovel **100** and the path of the work part of the shovel **100** up to a predetermined time in the future as inputs. The training data is a combination of the state of the target material of the shovel **100** at a

first time and a path of the work part of the shovel **100** from the first time to a second time after the first time as input data and the state of the target material at the second time as ground truth.
[0181] Note that the training data set for generating the trained model LM**1** may be generated only from the former log among the log acquired by the log acquisition part **2001** and the log output from the simulator part **2002**. In this case, the simulator part **2002** may be omitted. Similarly, the training data set for generating the trained model LM**1** may be generated only from the latter log among the log acquired by the log acquisition part **2001** and the log output from the simulator part **2002**. In this case, the operation log providing part **301** of the sensor group **300** and the shovel **100** may be omitted. In addition, the training data set for generating the trained model LM**1** may include a base training data set and a final adjustment (fine-tuning) training data set. In this case, since a large number of data is required, the base training data set may be generated based on the log output from the simulator part **2002**, and the final adjustment training data set may be generated based on the log acquired by the log acquisition part **2001**. Hereinafter, the same may be applied to the trained models LM**2** and LM**3**.
[0182] The training data generation part **2004**B generates training data for generating the trained model LM**2**. When a plurality of tasks are defined, the trained model LM**2** is generated for each of the plurality of tasks. The trained model LM**2** uses data of the state of the target material around the shovel **100** as an input, and infers one predetermined operation suitable for the state of the target material corresponding to the input data from among the plurality of predetermined operations used in the work.
[0183] The training data is, for example, a combination of the state of the target material before the performance of the predetermined operation of the shovel **100** as input data and the type of the predetermined operation performed by the shovel **100** thereafter as ground truth. The training data may further include the target shape (e.g., target construction surface) of the target material as input data. The training data generation part **2004**B may generate a training data set based on the log obtained by the log acquisition part **2001** when the shovel **100** performs the predetermined operation by a skilled person's operation. Thus, the trained model LM**2** can reproduce the way in which the skilled person selects a prescribed operation of the shovel **100**.
[0184] A training data generation part **2004**C generates training data for generating the trained model LM**3**. The trained model LM**3** is used for inferring a target path of a work part in a prescribed operation of the shovel **100** with data of the state of a target material around the shovel **100** as input. The trained model LM**3** is generated for each (type of) prescribed operation of the shovel **100**.
[0185] The trained model LM**3**, for example, infers an operation parameter that defines a target path of a work part in a prescribed operation of the shovel **100** based on the state of the target material before the performance of the prescribed operation of the shovel **100**. In this case, the training data is a combination of the state of the target material before the performance of the prescribed operation of the shovel **100** as input data and the operation parameter corresponding to the path of the work part when the shovel **100** performs the prescribed operation as ground truth. The trained model LM**3** may also infers a target path of a work part in a prescribed operation of the shovel **100** based on the state of the target material before the performance of the prescribed operation of the shovel **100**. In this case, the training data is a combination of the state of the target material before the performance of the prescribed operation of the shovel **100** as input data and the path of the work part when the shovel **100** performs the prescribed operation as ground truth. The training data may further include the target shape (e.g., target construction surface) of the target material as input data. In addition, the training data generation part **2004**C may generate a training data set based on a log obtained by the log acquisition part **2001** when the shovel **100** performs a predetermined operation by an expert. Thus, the trained model LM**3** can reproduce the operation of the shovel **100** by the expert.
[0186] The machine learning part **2005** causes the base learning model to perform machine

learning based on the training data set generated by the training data generation part **2004**, and generates the trained models LM**1** to LM**3**. The trained model (the base learning model) includes a neural network such as a DNN (Deep Neural Network), for example.

[0187] The machine learning part **2005** includes machine learning parts **2005**A to **2005**C.

[0188] The machine learning part **2005**A causes the base learning model M**1** to perform machine learning based on the training data set output from the training data generation part **2004**A. Thus, the machine learning part **2005**A can generate a trained model LM**1** capable of outputting (inferring) the state of the target material of the shovel **100** at a predetermined time in the future by taking data such as the current state of the target material of the shovel **100** and the target path of the work part of the shovel **100** up to a predetermined time in the future as inputs. Further, the machine learning part **2005**A may correct (additionally train) the trained model LM**1** so that an error between the inference result of the trained model LM**1** and the measurement result of an actual sensor **40** is reduced. In this case, the inference result of the trained model LM**1** and the data of the measurement result of the actual sensor **40** are uploaded from the shovel **100** to the information processing device **200**.

[0189] The machine learning part **2005**B causes the base learning model M**2** to perform machine learning based on the training data set output from the training data generation part **2004**B. Thus, the machine learning part **2005**B can generate the trained model LM**2** capable of outputting (inferring) one predetermined operation from among a plurality of predetermined operations corresponding to the target operation using data of the state of the target operation around the shovel **100** as an input. Further, the machine learning part **2005**B may generate the trained model LM**2** by performing reinforcement learning using the simulator part **2002**.

[0190] The machine learning part **2005**C causes the base learning model M**3** to perform machine learning based on the training data set output from the training data generation part **2004**C. Thus, the machine learning part **2005**C can generate the trained model LM**3** capable of outputting (inferring) the target path of the target operation in the predetermined operation of the shovel **100** using data of the state of the target operation around the shovel **100** as an input.

[0191] Trained models LM**1** and LM**2** output by the machine learning part **2005** are stored in the trained model storage part **2006**. When the trained model LM**1** is re-trained or additionally trained by the machine learning part **2005**A, the trained model LM**1** in the trained model storage part **2006** is updated. The same applies when the trained models LM**2** and LM**3** are re-trained or additionally trained by the machine learning parts **2005**B and **2005**C.

[0192] The distribution part **2007** distributes the data of the trained models LM**1** to LM**3** to the shovel **100**. For example, when the trained model LM**1** is generated or updated by the machine learning part **2005**A, the distribution part **2007** distributes the most recently generated or updated trained model LM**1** to the shovel **100**. The distribution part **2007** may also distribute the latest trained model LM**1** of the trained model storage part **2006** to the shovel **100** in response to a signal received from the shovel **100** requesting the distribution of the trained model LM**1**. The same may be applied to the trained models LM**2** and LM**3**.

[0193] The work support part **302** is a functional part for supporting the work of the shovel **100** operating by the autonomous operation function.

[0194] The work support part **302** includes a trained model storage part **302**A, a target material state prediction part **302**B, an operation planning part **302**C, a target path generation part **302**D, and an operation control part **302**E.

[0195] The trained model storage part **302**A stores trained models LM**1** and LM**2** distributed from the information processing device **200** received and through the communication device **60**.

[0196] The target material state prediction part **302**B predicts the state of the target material of the shovel **100** at a predetermined time in the future based on the current state of the target material around the shovel **100** and the target path of the work part of the shovel **100** up to a predetermined time. Specifically, the target material state prediction part **302**B predicts the state of the target material of the shovel **100** at a future time using the trained model LM**1**.

[0197] The current state of the target material around the shovel **100** is acquired, for example, based on the output of the sensor **40**. The current state of the target material around the shovel **100** may be acquired based on the output of the sensors S**7** to S**9** instead of or in addition to the sensor **40**. This is because the controller **30** can estimate the reaction force from the ground acting on the bucket **6** from the output of the sensors S**7** to S**9**, and can estimate the state (shape and characteristics) of the earth of the target material from the estimated result of the reaction force. The current state of the target material around the shovel **100** may be a prediction result of the state of the target material output by the target material state prediction part **302**B itself at a processing timing earlier than the current time. In this case, for example, the target material state prediction part **302**B predicts the state of the target material based on the initial state of the target material at the beginning of the work, and subsequently uses the prediction result as the current state of the target material. The initial state of the target material at the beginning of the work may be provided from the outside of the shovel **100**, or may be previously defined as a fixed state, such as a plane at the same height as the ground on which the lower traveling body **1** of the shovel **100** is grounded.

[0198] The predetermined time point is, for example, the timing (time t.sub.b) of the start of the predetermined operation that can be modified immediately. A modification of the predetermined operation means, for example, the modification of the type of the predetermined operation that is scheduled to be performed to another type, such as the modification of a future predetermined in operation an undetermined or tentatively determined state or an excavation operation that is scheduled to be performed to a slope pulling operation. The predetermined time point may be the timing (time t.sub.s) at which the path of the bucket **6** and the type of the predetermined operation can be modified, considering the delay time τ.sub.s in the process. The delay time τ.sub.s includes, for example, the calculation time for the controller **30** to generate the target path of the bucket **6** or to determine the predetermined operation of the shovel **100**, and the interface time to transfer the calculation result to the control part.

[0199] The time t.sub.s is calculated by the following equation (1) using the current time t.sub.1 and the delay time τ.sub.s.

[00001] [Math1]

$$t_s = t_1 + \;_s \quad (1)$$

[0200] The delay time τ.sub.s may be a fixed value or a variable value. In the former case, the fixed value is previously defined as the maximum value of the delay time that can be assumed depending on, for example, the processing state of the controller **30**. In the latter case, the delay time τ.sub.s is varied according to a predetermined rule depending on, for example, the processing state of the controller **30** such as the CPU load state.

[0201] The time t.sub.b corresponds to a start time of the next predetermined operation of the predetermined operation being performed at the time t.sub.s.

[0202] For example, as shown in FIG. **7**, when the time t.sub.s is before the start time of the next operation B of the current operation A of the shovel **100**, the time t.sub.b corresponds to the start time of the next operation B of the current operation A.

[0203] Conversely, as shown in FIG. **8**, when the time t.sub.s is after the start time of the next operation B of the current operation A of the shovel **100**, the controller **30** cannot modify the next operation B to another type of predetermined operation due to the delay time τ.sub.s. Therefore, in this case, the time t.sub.b corresponds to the start time of the next operation C of the operation B being performed at the time t.sub.s.

[0204] The operation planning part **302**C plans (determines) the predetermined operation (type) to be performed by the shovel **100** from the time t.sub.b based on the prediction result (prediction result of the state of the target material at the time t.sub.b) by the target material state prediction part **302**B.

[0205] Thus, the controller **30** can determine the predetermined operation type to be performed by the shovel **100** in accordance with the predicted state of the target material in the future (time t.sub.b). Therefore, for example, it is not necessary to stop the operation of the shovel **100** to some extent, as in the case of determining the subsequent predetermined operation based on the actual state of the target material at the time of completion of the predetermined operation immediately before the time t.sub.b. Therefore, the controller **30** can improve the work efficiency of the shovel **100**.
[0206] The operation planning part **302**C may adjust the processing timing so as to plan the next predetermined operation after the predetermined operation currently being performed by the shovel **100**. For example, when the remaining time until the end time of the predetermined operation currently being performed by the shovel **100** is not less than the delay time τ.sub.s, the operation planning part **302**C plans the next predetermined operation of the shovel **100**.
[0207] For example, the operation planning part **302**C determines the predetermined operation (type) to be started at the time t.sub.b based on the prediction result of the state of the target material at the time t.sub.b by using a rule-based method.
[0208] For example, a plurality of predetermined operations that can be performed are defined for each of the target materials, and transition conditions for each of the previously defined operations that can transition are specified in advance for each of the predetermined operations. The predetermined operations that can transition can include the same predetermined operation. This is because the same predetermined operation may be repeated. Then, the operation planning part **302**C determines the predetermined operation (type of operation) to be started at time t.sub.b based on the success or failure of a plurality of transition conditions starting from the predetermined to be performed operation immediately before time t.sub.b.
[0209] For example, as shown in FIG. **9**, in the slope work of this example, the excavation operation ST**1-1**, the earth removing operation ST**1-2**, and the slope pulling operation ST**1-3** are defined as the plurality of predetermined operations that can be performed.
[0210] The predetermined operations that can be transitioned from a standby state ST**1-0** corresponding to before the work start and after the work completion are the excavation operation ST**1-1** and the slope pulling operation ST**1-3**, and the transition conditions SC**1-01** and SC**1-03** are defined respectively. The transition conditions SC**1-01** and SC**1-03** are mutually opposing conditions. For example, if a difference between the shape of the target material and the target shape at the beginning of the work is not more than the predetermined standard, the transition condition SC**1-03** is established, and if it exceeds the predetermined standard, the transition condition SC**1-01** is established. The predetermined operation to be performed at the beginning of the work is determined to be the predetermined operation corresponding to one of the transition conditions SC**1-01** and SC**1-03**.
[0211] The predetermined operation that can transition from the excavation operation ST**1-1** is the earth removing operation ST**1-2**, and the transition condition SC**1-12** is defined. That is, when the predetermined operation performed immediately before time t.sub.b is the excavation operation ST**1-1**, the transition condition SC**1-12** is always established, and the predetermined operation performed from time t.sub.b is unambiguously determined to be the earth removing operation ST**1-2**.
[0212] The predetermined operations that can transition from the earth removing operation ST**1-2** are the excavation operation ST**1-1** and the slope pulling operation ST**1-3**, and the transition conditions SC**1-21** and SC**1-23** are defined respectively. The transition conditions SC**1-21** and SC**1-23** are mutually opposing conditions. For example, when the difference between the predicted result of the shape of the target material and the target shape at time t.sub.b is not more than the predetermined standard, the transition condition SC**1-23** is established, and when it exceeds the predetermined standard, the transition condition SC**1-21** is established. When the predetermined operation to be performed immediately before time t.sub.b is the earth removing operation ST**1-2**,

the predetermined operation to be performed from time t.sub.b is determined to be the predetermined operation corresponding to one of the transition conditions SC**1-21** and SC**1-23**.
[0213] The predetermined operations that can transition from the slope pulling operation ST**1-3** are the excavation operation ST**1-1** and the slope pulling operation ST**1-3**, and the transition conditions SC**1-31** and SC**1-33** are defined respectively. The transition conditions SC**1-31** and SC**1-33** are mutually opposite. If the predetermined operation performed immediately before time t.sub.b is the slope pulling operation ST**1-3**, the predetermined operation performed from time t.sub.b is determined to be the predetermined operation corresponding to one of the transition conditions SC**1-31** and SC**1-33**.
[0214] If the work completion condition indicating that the work has been completed is satisfied at time t.sub.b, the controller **30** shifts to the standby state without performing the predetermined operation determined in advance by the operation planning part **302**C (see a dashed arrow in the figure). For example, the work completion condition is that the difference between the shape of the target material and the target shape just before time t.sub.b is so small that it can be considered zero.
[0215] Further, the operation planning part **302**C may use the trained model LM**2** to determine the predetermined operation (type of operation) to be started by the shovel **100** at time t.sub.b based on the prediction result of the state of the target material at time t.sub.b.
[0216] For example, as shown in FIG. **10**, in the case of the ground leveling operation, there are many predetermined operations that can be performed and the combination of transition destinations for each predetermined operation is complicated. As a result, there is a possibility that the transition condition starting from a certain predetermined operation cannot be properly set. However, by using the trained model LM**2**, even in the case of the ground leveling operation, where there are many predetermined operations that can be performed and the combination of transition destinations for each predetermined operation is complicated, the operation planning part **302**C can more appropriately determine the predetermined operation to be started by the shovel **100** at time t.sub.b.
[0217] When the operation planning part **302**C can determine the predetermined operation to be performed by the shovel **100** from time t.sub.b only by the rule-based method, the training data generation part **2004**B and the machine learning part **2005**B are omitted.
[0218] The target path generation part **302**D generates the target path of the work part in the predetermined operation of the shovel **100** based on the state of the target material around the shovel **100**. The predetermined operation at this time is the type of predetermined operation determined by the operation planning part **302**C.
[0219] For example, the target path generation part **302**D generates the target path of the work part after time t.sub.s using the trained model LM**3** based on the prediction result (predicted state of the target material at time t.sub.s, t.sub.b) of the target material state prediction part **302**B. Thus, for example, the controller **30** can modify the target path of the work part in the predetermined operation of the shovel **100** in accordance with the state (prediction result) of the target material during performance of the predetermined operation. Therefore, the shovel **100** can progress the work of the shovel **100** more appropriately and efficiently according to the change of the state of the target material.
[0220] Specifically, the target path generation part **302**D may generate the target path of the work part in the predetermined operation performed at time t.sub.s and the target path of the work part in the predetermined operation started from time t.sub.b.
[0221] The target path generation part **302**D may generate the target path of the work part of the shovel **100** in accordance with the state (prediction result) of the target material around the shovel **100** by applying any known technique instead of the trained model LM**3**. In this case, the training data generation part **2004**C and the machine learning part **2005**C may be omitted. For example, the target path generation part **302**D may generate data of the target path of the work part of the shovel

100 by MPC (Model Predictive Control) based on the prediction result (predicted state of the target material at time t.sub.s, t.sub.b) of the target material state prediction part **302**B. In addition, the target path generation part **302**D may generate data of the target path of the work part of the shovel **100** by optimizing the previously defined reference path of the work part of the shovel **100** based on data concerning the characteristics of earth given in advance.
[0222] The operation control part **302**E causes the shovel **100** to perform a predetermined operation so that the predetermined part of the shovel **100** moves along the target path generated by the target path generation part **302**D. Specifically, the operation control part **302**E causes the shovel **100** to perform a predetermined operation so that the work part of the shovel **100** moves along the target path by controlling the hydraulic controlling valve **31** while determining the position of the work part from the outputs of the sensors S**1** to S**5**. Thus, the shovel **100** can autonomously advance the work while performing the predetermined operation according to the shape of the target material.
[0223] Thus, in this example, the controller **30** determines a future predetermined operation of the shovel **100** from among a plurality of predetermined operations corresponding to the target work according to the performance status of the predetermined operation of the working machine. Specifically, the controller **30** may predict the future state of the target material according to the performance status of the operation of the working machine. Then, the controller **30** may determine a future predetermined operation of the shovel **100** from among a plurality of predetermined operations corresponding to the target work based on the prediction result of the future state of the target material. Thus, the controller **30** can determine in advance the operation of the shovel **100** to be performed in accordance with the future state of the target material. Therefore, idle time when the predetermined operation of the shovel **100** is completed and the next predetermined operation is performed can be reduced, and the work efficiency of the shovel **100** can be improved.
[0224] Moreover, in this example, after determining the future predetermined operation of the shovel **100**, the controller **30** generates a target path of the work part corresponding to the determined predetermined operation of the shovel **100**. Thus, the controller **30** can hierarchically determine the future predetermined operation of the shovel **100** and generate a target path of the work part in the future predetermined operation of the shovel **100**. Therefore, for example, when the operation plan of the shovel **100** and the path plan of the work part of the shovel **100** are performed in parallel, it is possible to control against an occurrence of a situation in which the operation plan and the path plan cannot be performed in a realistic time as a result of enormous conditions and parameters.
[0225] Some or all of the functions of the target material state prediction part **302**B, the operation planning part **302**C, the target path generation part **302**D, and the operation control part **302**E may be transferred to the information processing device **200**. Thus, the processing load of the shovel **100** can be reduced for the processing related to the generation of the target path of the work part of the shovel **100** and the processing related to the control of the operation of the shovel **100**.
Second Example
[0226] FIG. **11** is a functional block diagram illustrating a second example of the functional configuration of the operation support system SYS.
[0227] Hereinafter, the same reference numerals are assigned to the same or corresponding configurations as in the first example described above, and the description will be made mainly on the different parts from the first example described above.
[0228] The shovel **100** includes a support device **150** as in the first example described above. In this example, the support device **150** supports a user who operates the semi-automatic shovel **100** and performs work.
[0229] As shown in FIG. **11**, in this example, as in the first example described above, the support device **150** includes a controller **30**, a hydraulic controlling valve **31**, a sensor **40**, an output device **50**, and sensors S**1** to S**9**. When the shovel **100** is remotely operated, the support device **150** may include a communication device **60** instead of or in addition to the input device **52**.

[0230] As in the first example described above, the controller **30** includes, as functional parts, an operation log providing part **301** and a work support part **302**.
[0231] As in the first example described above, the operation log providing part **301** includes, as functional parts, an operation log recording part **301**A, an operation log storage part **301**B, and an operation log transmission part **301**C.
[0232] As in the first example described above, the information processing device **200** includes, as functional parts, a log acquisition part **2001**, a simulator part **2002**, a log storage part **2003**, a training data generation part **2004**, a machine learning part **2005**, a trained model storage part **2006**, and a distribution part **2007**.
[0233] The work support part **302** is a functional part for supporting a user who operates the semi-automatic shovel **100** and performs work.
[0234] As in the first example described above, the work support part **302** includes a trained model storage part **302**A, a target material state prediction part **302**B, an operation planning part **302**C, a target path generation part **302**D, and an operation control part **302**E. Unlike the first example described above, the work support part **302** includes an operation proposal part **302**F.
[0235] The operation proposal part **302**F proposes predetermined operation (type) of the shovel **100** from time t.sub.b determined (planned) by the operation planning part **302**C to the user through the output device **50** and the remote control support device **400**.
[0236] In response to the proposal of the predetermined operation of the shovel **100** by the operation proposal part **302**F, the user selects a predetermined operation (type) to be performed by the shovel **100** starting from time t.sub.b from among a plurality of predetermined operations corresponding to the current work. At this time, the user selects a predetermined operation to be performed by the shovel **100** starting from time t.sub.b using the input device **52** and the remote control support device **400**, for example. The selection result by the user is input to the target path generation part **302**D through the input device **52** and the communication device **60**.
[0237] The target path generation part **302**D generates the target path of the work part after time t.sub.s by using the trained model LM**3** based on the prediction result of the target material state prediction part **302**B (the prediction result of the state of the target material at time t.sub.s).
[0238] Specifically, the target path generation part **302**D may generate the target path of the work part in the predetermined operation performed at time t.sub.s and the target path of the work part in the predetermined operation started at time t.sub.b. In this case, the predetermined operation started at time t.sub.b is the predetermined operation of the shovel **100** corresponding to the selection result by the user.
[0239] Thus, in this example, the controller **30** determines the future predetermined operation of the shovel **100** from among a plurality of predetermined operations corresponding to the target operation based on the prediction result of the state of the target material in the future, and proposes it to the user through the output device **50** and the remote control support device **400**. Thus, the user can determine the type of a recommended predetermined operation of the shovel **100** based on the prediction result of the state of the target material in the future before the completion of the predetermined operation immediately before the recommended predetermined operation of the shovel **100**. Therefore, idle time when the predetermined operation of the shovel **100** is completed and the next predetermined operation is performed can be reduced, and the work efficiency of the shovel **100** can be improved.
[0240] The functions of the operation planning part **302**C and the operation proposal part **302**F may be adopted in a manually operated shovel **100**, whose operations are entirely controlled by the operator. In this case, the controller **30** may predict the path of the work part in the predetermined operation of the shovel **100** based on the history of the operation contents of the operator or the output of the sensors S**1** to S**9**, and may predict the future state of the target material based on the predicted path. When the shovel **100** is operated remotely, some or all of the functions of the trained model storage part **302**A, the target material state prediction part **302**B, the operation

planning part **302**C, the target path generation part **302**D, the operation control part **302**E, and the operation proposal part **302**F may be provided in the remote control support device **400**. In addition, some or all of the functions of the target material state prediction part **302**B, the operation planning part **302**C, the target path generation part **302**D, the operation control part **302**E, and the operation may be transferred to the proposal part **302**F information processing device **200**. Thus, the processing load of the shovel **100** and the remote control support device **400** can be reduced in the processing related to the generation of the target path of the work part of the shovel **100** and the processing related to the control of the operation of the shovel **100**.

[Specific Example of Processing Related to Autonomous Operation of Shovel]

[0241] Next, a specific example of processing related to the autonomous operation of the shovel **100** will be described with reference to FIGS. **12** to **17**.

[0242] In this example, the description proceeds assuming the functional configuration of the operation support system SYS shown in FIG. **6**.

<Process Related to Start of Autonomous Operation of Shovel>

[0243] FIG. **12** is a flowchart schematically illustrating an example of processing related to a start of autonomous operation of the shovel **100**. This flowchart is performed when a predetermined input related to the start of autonomous operation is made by the user through the input device **52**, the remote control support device **400**, or the remote monitoring support device.

[0244] In a step S**102**, the controller **30** selects a predetermined operation to be performed at the beginning of autonomous operation of the shovel **100** in response to a predetermined input from the user through the input device **52**, the remote control support device **400**, or the remote monitoring support device. For example, the controller **30** selects one predetermined operation from among a plurality of predetermined operations defined for the target work such as excavation work, leveling work, and slope work. Further, prior to the step S**102**, the controller **30** may select one operation from among a plurality of operations such as excavation work, leveling work, and slope work in response to a predetermined input from the user through the input device **52**, the remote control support device **400**, or the remote monitoring support device.

[0245] When the processing of the step S**102** is completed, the controller **30** proceeds to a step S**104**.

[0246] In the step S**104**, the controller **30** acquires data representing the state (shape and characteristic) of the earth to be worked, based on the output of the sensor **40**.

[0247] When the processing of the step S**104** is completed, the controller **30** proceeds to a step S**106**.

[0248] In the step S**106**, the controller **30** (target path generation part **302**D) generates the target path of the bucket **6** for the predetermined operation of the shovel **100** selected in the step S**102** at the beginning of the work, based on the data acquired in the step S**104**. For example, the target path generation part **302**D generates the target path of the work part of the shovel **100** in the same manner as in a step S**208** described later.

[0249] When the processing of the step S**106** is completed, the controller **30** proceeds to the step S**108**.

[0250] In the step S**108**, the controller **30** notifies the user that autonomous operation is enabled. For example, the controller **30** notifies the user inside the cabin **10** or the user near the shovel **100** through the output device **50**. The controller **30** may also notify the user using the remote control support device **400** or the remote monitoring support device by transmitting the notification signal to the remote control support device **400** or the remote monitoring support device through the communication device **60**.

[0251] When the processing in the step S**108** is completed, the controller **30** proceeds to a step S**110**.

[0252] In the step S**110**, the controller **30** starts the autonomous operation of the shovel **100** in response to an instruction from the user received through the input device **52**, the remote control support device **400**, or the remote monitoring support device.

[0253] When the processing of the step S**110** is completed, the controller **30** ends processing of the present flowchart.

[0254] Thus, the controller **30** can start the autonomous operation of the shovel **100** in response to a predetermined input from the user.

<Process Related to Bucket Path Generation (Main Flow)>

[0255] FIG. **13** is a main flowchart schematically illustrating an example of processing related to bucket path generation. FIG. **14** is a drawing illustrating an example of an observation target area TA.

[0256] This flowchart is repeatedly performed every predetermined control cycle after the start of the autonomous operation of the shovel **100**.

[0257] In a step S**202**, the controller **30** acquires the future times t.sub.s and t.sub.b as references for the operation plan of the shovel **100** and the path generation.

[0258] When the processing in the step S**202** is completed, the controller **30** proceeds to a step S**204**.

[0259] In the step S**204**, the target material state prediction part **302**B predicts the state of the earth of the target material (ground) at times t.sub.s and t.sub.b. Specifically, the target material state prediction part **302**B predicts the state of the earth of the target material at a modifiable start time t.sub.s based on the state of the earth of the target material at the current time t.sub.1 and the target path of the bucket **6** from the current time t.sub.1 to the time t.sub.s. Similarly, the target material state prediction part **302**B predicts the state of the earth of the target material at time t.sub.b based on the state of the earth of the target material at the current time t.sub.1 and the target path of the bucket **6** from the current time t.sub.1 to the time t.sub.b. The data of the target path of the bucket **6** used in this step is obtained in the process of a step S**208** in the previous flow chart or in the process of the step S**106** in FIG. **8**.

[0260] For example, as shown in FIG. **14**, the observation target area TA around the shovel **100** is divided into grid units of a predetermined number N. The observation target area TA is an area around the shovel **100** in which the target material state prediction part **302**B acquires data indicating the state of the earth. In this example, at time t, a shape h.sup.t of the earth and a characteristic κ.sup.t of the earth of each grid unit i (i=1 to N) of the observation target area TA are defined.

[0261] For example, the shape h.sup.t of the earth and the characteristic κ.sup.t of the earth of the work target of the shovel **100** for each grid unit i of the observation target area TA at the current time t.sub.1 are expressed by the following equations (2) and (3).

[Math2]

[00002] $$h^t = [h_1^t, h_2^t, \dots, h_N^t]^T \quad (2) \qquad {}^t = [\ {}_1^t,\ {}_2^t,\ \dots,\ {}_N^t]^T \quad (3)$$

[0262] For example, the target path X.sub.t1:ts of the bucket **6** from the current time t.sub.1 to the modifiable start time t.sub.s and the target path X.sub.t1:tb of the bucket **6** from the current time t.sub.1 to the time t.sub.b are expressed by the following equations (4) and (5).

[Math3]

[00003] $$X_{t1:\mathrm{ts}} = \{X_{t1}, X_{t1+1}, \dots, X_{\mathrm{ts}}\} \quad (4) \qquad X_{t1:\mathrm{tb}} = \{X_{t1}, X_{t1+1}, \dots, X_{\mathrm{tb}}\} \quad (5)$$

[0263] That is, in this example, the target path X.sub.t1:ts of the bucket **6** is defined as a set of positions X.sub.t of the bucket **6** at each time t which are expressed discretely. For example, the position X.sub.t of the bucket **6** at time t is expressed by the following equation (6) as a set of the attitude θ.sub.1, t of the boom **4**, the attitude θ.sub.2, t of the arm **5**, the attitude θ.sub.3, t of the bucket **6**, and the attitude θ.sub.4, t of the upper swivel body **3**.

[Math4]

[00004] $$X_t = [\ {}_{1,t},\ {}_{2,t},\ {}_{3,t},\ {}_{4,t}]^T \quad (6)$$

[0264] The attitude θ.sub.1, t of the boom **4** is information representing, for example, the position (rod position) of the boom cylinder **7**. The attitude θ.sub.1, t of the boom **4** may be information representing the attitude angle of the boom **4**. The data of the attitude θ.sub.1, t of the boom **4** is acquired based on the output of the sensor S**7**.
[0265] The attitude θ.sub.2, t of the arm **5** is, for example, information representing the position (rod position) of the arm cylinder **8**. The attitude θ.sub.2, t of the arm **5** may be information representing the attitude angle of the arm **5**. The data of the attitude θ.sub.2, t of the arm **5** is acquired based on the output of the sensor S**8**.
[0266] The attitude θ.sub.3, t of the bucket **6** is, for example, information representing the position (rod position) of the bucket cylinder **9**. The attitude θ.sub.3, t of the bucket **6** may be information representing the attitude angle of the bucket **6**. The data of the attitude θ.sub.3, t of the bucket **6** is acquired based on the output of the sensor S**9**.
[0267] The attitude θ.sub.4, t of the upper swivel body **3** is, for example, information representing the swivel angle of the upper swivel body **3**. The data of the attitude θ.sub.4, t of the upper swivel body **3** is acquired based on the output of the sensor S**4** or the sensor S**5**.
[0268] In addition, the position X.sub.t of the bucket **6** may include information on the velocity, acceleration, or jerk and the like of each of the boom **4**, the arm **5**, and the bucket **6**.
[0269] For example, based on the shape h.sup.t of the earth and the characteristic κ.sup.t of the earth at the current time t.sub.1 for each grid unit i of the observation target area TA, and the target path X.sub.t1:ts of the bucket **6** from the current time t.sub.1 to the time t.sub.s, the target material state prediction part **302**B predicts the shape h.sup.ts of the earth and the earth kts of the earth at the time t.sub.s by using a function g corresponding to the trained model LM**1**. The shape h.sup.ts of the earth and the characteristic kts of the earth at the time t.sub.s is expressed by the following equation (7).

[Math5]

[00005] $$(h^{ts},\ {}^{ts}) = g(h^{t1},\ {}^{t1}, X_{t1:ts}) \quad (7)$$

[0270] Similarly, based on the shape h.sup.t of the earth and the characteristic κ.sup.t of the earth at the current time t.sub.1 for each grid unit i of the observation target area TA, and the target path X.sub.t1:tb of the bucket **6** from the current time t.sub.1 to the time t.sub.b, the target material state prediction part **302**B predicts the shape h.sup.tb of the earth and the characteristic κ.sup.tb of the earth at the time t.sub.b by using a function g. The shape h.sup.tb of the earth and the characteristic κ.sup.tb of the earth at the time t.sub.b is expressed by the following equation (8).

[Math6]

[00006] $$(h^{tb},\ {}^{tb}) = g(h^{t1},\ {}^{t1}, X_{t1:th}) \quad (8)$$

[0271] The function g is based on, for example, a DNN.
[0272] When the processing in the step S**204** is completed, the controller **30** proceeds to a step S**206**.
[0273] In the step S**206**, the operation planning part **302**C determines the next predetermined operation (type) v.sup.k that can be modified from among a plurality of predetermined operations corresponding to the current operation based on the current predetermined operation (type) v.sup.k-1 and the state of the earth predicted in the step S**204**.
[0274] For example, the operation planning part **302**C determines the next predetermined operation (type) v.sup.k that can be modified from among a plurality of predetermined operations corresponding to the current operation using the function f corresponding to the trained model LM**2**. The predetermined operation v.sup.k is expressed by the following equation (9) using the function f.

[Math7]

[00007] $$v^{k} = f(v^{k-1}, h^{tb},\ {}^{tb}) \quad (9)$$

[0275] The predetermined operations v.sup.k and v.sup.k-1 are expressed by, for example, a one-hot vector.
[0276] The function f may be defined in the rule base instead of being given by the trained model LM**2** as described above.
[0277] When the processing of the step S**206** is completed, the controller **30** proceeds to a step S**208**.
[0278] In the step S**208**, the target path generation part **302**D generates the target path of the bucket **6** in the predetermined operation of the shovel **100** after the time t.sub.s, based on the prediction result (the state of the earth at the time t.sub.s) of the process in step S**204**. Specifically, the target path generation part **302**D generates the target path of the bucket **6** in the predetermined operation of the shovel **100** from the time t.sub.s to the time t.sub.b, and the target path of the bucket **6** in the predetermined operation of the shovel **100** after the time t.sub.b until the predetermined timing.
[0279] When the process in the step S**208** is completed, the controller **30** proceeds to a step S**210**.
[0280] In the step S**210**, the target path generation part **302**D writes the data of the target path of the bucket **6** generated in the step S**208** to a predetermined storage area (address) of the memory device **30**B.
[0281] Thus, the operation control part **302**E can access the latest data of the target path of the shovel **100** by accessing the predetermined address of the memory device **30**B.
[0282] When the process in the step S**210** is completed, the controller **30** ends the process of the present flowchart.
[0283] Thus, in this example, the controller **30** predicts a future shape of the earth, and determines the predetermined future operation of the shovel **100** based on the future shape of the earth. Thus, the controller **30** can generate the target path of the bucket **6** based on the predicted future shape of the earth and the determined predetermined future operation.
<Process Related to Target Path Generation (Sub-Flow)>
[0284] FIG. **15** is a sub-flowchart schematically illustrating an example of processing related to path generation of the bucket **6**. FIG. **16** is a drawing illustrating an example of cost conditions and operation parameters corresponding to a plurality of sections of a shovel **100** excavation operation.
[0285] The sub-flowchart of FIG. **15** corresponds to the process of the step S**208** in FIG. **13**.
[0286] As shown in FIG. **15**, in a step S**302**, the target path generation part **302**D selects a constraint function related to the path of the bucket **6** corresponding to (type of) a predetermined operation of the shovel **100**. At this time, the predetermined operation of the shovel **100** is a predetermined operation of the shovel **100** performed at time t.sub.s and a predetermined operation of the shovel **100** starting from time t.sub.b. Specifically, for each predetermined operation, the target path generation part **302**D selects a constraint function related to the path of the bucket **6** corresponding to the predetermined operation.
[0287] The constraint function (constraint condition) includes, for example, constraints related to the moving range, speed, and acceleration of the boom cylinder **7**, the arm cylinder **8**, and the bucket cylinder **9**. Further, the constraint function (constraint condition) may include a constraint condition for avoiding collision between an obstacle around the shovel **100** and the bucket **6**. The obstacle around the shovel **100** includes, for example, a person, a working vehicle, another working machine, a feature (e.g., fence, utility pole), and the like, and can be recognized based on the output of the sensor **40**.
[0288] When the processing of the step S**302** is completed, the controller **30** proceeds to a step S**304**.
[0289] In the step S**304**, the target path generation part **302**D selects an objective function (cost function) related to the path of the bucket **6** corresponding to (type of) the predetermined operation of the shovel **100**. At this time, the predetermined operation of the shovel **100** is the predetermined operation of the shovel **100** performed at time t.sub.s and the predetermined operation of the shovel **100** starting from time t.sub.b, as in the case of step S**302**. Specifically, for each predetermined

operation, the target path generation part **302**D selects a cost function related to the path of the bucket **6** corresponding to the predetermined operation.

[0290] For example, as shown in FIG. **16**, the excavation operation as the predetermined operation of the shovel **100** is divided into the operation sections of approach, penetration, horizontal excavation, and scooping up.

[0291] "Approach" is an operation section in which the bucket **6** is brought close to the ground in order to penetrate the ground. "Penetration" is an operation section in which the cutting edge of the bucket **6** is brought into contact with the ground after the operation section of approach and the bucket **6** is penetrated to a certain depth of the ground. "Horizontal excavation" is an operation section in which the bucket is 6 moved approximately horizontally after the operation section of penetration. "Scooping up" is an operation section in which earth are stored in the bucket **6** after horizontal excavation and the earth are scooped up on the ground.

[0292] In this example, cost functions related to the speed and acceleration of the bucket **6**, the travel time of the bucket **6**, and the like are defined over all the operation sections from "approach" to "scooping up".

[0293] In addition, cost functions related to the position of the cutting edge of the bucket **6**, the angle of the cutting edge to a predetermined reference (e.g., a horizontal plane), and the path of the cutting edge are defined at the end of one operation section among "approach", "penetration", "horizontal excavation", and "scooping up", and in one operation section among the above.

[0294] When the processing of the step S**304** is completed, the controller **30** proceeds to a step S**306**.

[0295] In the step S**306**, the target path generation part **302**D estimates an operation parameter that defines the target path of the bucket **6** in a predetermined operation of the shovel **100** using the trained model LM**3** based on the prediction result (state of the earth of the target material at time t.sub.s, t.sub.b) in the step S**302**.

[0296] For example, as shown in FIG. **16**, operation parameters q.sub.1 to q.sub.4 that define the position of the cutting edge of the bucket **6**, and operation parameters β.sub.12, β.sub.23, and ρ.sub.4 that define the angle of the cutting edge of the bucket **6** to a predetermined reference are defined.

[0297] The operation parameter q.sub.1 is an operation parameter representing the position of the cutting edge of the bucket **6** at the end of "approach" and at the beginning of "penetration". In this example, at the end of "approach" and at the beginning of "penetration", a cost function corresponding to the condition for determining that the position of the cutting edge of the bucket **6** matches the position of the operation parameter q.sub.1 is defined.

[0298] The operation parameter q.sub.2 is an operation parameter representing the position of the cutting edge of the bucket **6** at the end of "penetration" and at the beginning of "horizontal excavation". In this example, at the end of "penetration" and at the beginning of "horizontal excavation", a cost function corresponding to the condition for determining that the position of the cutting edge of the bucket **6** matches the position of the operation parameter q.sub.2 is defined.

[0299] In this example, a cost function corresponding to the condition for determining that the position (path) of the cutting edge is on the straight line defined by the operation parameters q.sub.1 and q.sub.2 in the operation section of "penetration" is defined.

[0300] The operation parameter q.sub.3 is an operation parameter representing the position of the cutting edge of the bucket **6** at the end of "horizontal excavation" and at the beginning of "scooping up". In this example, the cost function corresponding to the condition for determining that the position of the edge of the bucket **6** and the position of the operation parameter q.sub.3 match at the end of "horizontal excavation" and at the beginning of "scooping up" is defined.

[0301] In this example, the cost function corresponding to the condition for determining that the position (path) of the edge is on the straight line defined by the operation parameters q.sub.2 and q.sub.3 in the operation section of "horizontal excavation" is defined.

[0302] The operation parameter q.sub.4 is an operation parameter representing the position of the edge of the bucket **6** at the end of "scooping up". In this example, the cost function corresponding to the condition for determining that the position of the edge of the bucket **6** and the position of the operation parameter q.sub.4 match at the end of "scooping up" is defined.
[0303] The operation parameter ρ.sub.12 is an operation parameter representing the angle of the edge of the bucket **6** with respect to the predetermined reference in the operation section of "penetration". In this example, the constraint function and cost function corresponding to the condition for determining that the angle of the edge of the bucket **6** with respect to the predetermined reference and the angle of the operation parameter ρ.sub.12 match in the operation section of "penetration" are defined.
[0304] The operation parameter ρ.sub.23 is an operation parameter representing the angle of the cutting edge of the bucket **6** with respect to a predetermined reference in the operation section of "horizontal excavation". In this example, in the operation section of "horizontal excavation", the constraint function and the cost function, corresponding to the condition for determining that the angle of the cutting edge of the bucket **6** with respect to a predetermined reference matches the angle of the operation parameter ρ.sub.23, are defined.
[0305] The operation parameter ρ.sub.4 is an operation parameter representing the angle of the cutting edge of the bucket **6** with respect to a predetermined reference at the end of "scooping up". In this example, the constraint function and the cost function corresponding to the condition for determining that the angle of the cutting edge of the bucket **6** with respect to a predetermined reference matches the angle of the operation parameter ρ.sub.4 at the end of "scooping up" are defined.
[0306] When the processing in the step S**306** is completed, the controller **30** proceeds to a step S**308**.
[0307] In the step S**308**, the target path generation part **302**D calculates the target path of the bucket **6** in the predetermined operation of the shovel **100** based on the constraint function and the objective function selected in the steps S**302** and S**304**, and the operation parameter of the estimation result in the step S**306**. Specifically, the target path generation part **302**D calculates the target path of the bucket **6** in the predetermined operation of the shovel **100** by solving a constrained nonlinear optimization problem defined by the constraint function and the objective function using a predetermined solver.
[0308] When the processing of the step S**308** is completed, the processing of the present sub-flowchart is terminated.
[0309] Thus, in this example, the controller **30** can generate the target path of the shovel **100** by using the constraint function and the objective function corresponding to the predetermined operation of the shovel **100**.
[0310] In this example, the path of the bucket **6** is represented by a relatively small number of operation parameters. Then, the controller **30** can infer the operation parameters which define the path of the bucket **6** by using the trained model LM**3** based on the prediction result of the state of the earth of the future target material.
<Processing Related to Operation Control of Shovel>
[0311] FIG. **17** is a flowchart schematically illustrating an example of processing related to operation control of the shovel **100**.
[0312] This flowchart is repeatedly performed for each predetermined processing cycle during performance of the autonomous operation of the shovel **100**, for example.
[0313] As shown in FIG. **17**, in a step S**402**, the operation control part **302**E reads the latest data representing the target path of the bucket **6** in the predetermined operation of the shovel **100** from the predetermined address of the memory device **30**B. This data is the data registered in the step S**210** of FIG. **13**.
[0314] When the processing in the step S**402** is completed, the controller **30** proceeds to a step

S**404**.
[0315] In the step S**404**, the operation control part **302**E controls the operation of the shovel **100** based on the data of the target path of the bucket **6** in the predetermined operation of the shovel **100** read in the step S**402**. Specifically, the operation control part **302**E controls the operation of the shovel **100** while outputting a control command to the hydraulic controlling valve **31** so that the bucket **6** moves along the target path corresponding to the data read in the step S**402**.
[0316] When the process of the step S**404** is completed, the process of the present flowchart ends.
[0317] Thus, in this example, the controller **30** can control the operation of the shovel **100** so as to move along the target path of the predetermined operation of the shovel **100**.
[Specific Example of Trained Model Generation Method]
[0318] Next, a specific example of a method for generating a trained model LM**1** will be described.
[0319] In this example, a method for generating a function g corresponding to the trained model LM**1** used in the process of FIG. **13** will be described.
[0320] Training data c.sup.j (j=1 to L (an integer of 2 or more)) of a training data set D generated by the training data generation part **2004**A is expressed, for example, by the following equation (10).

[Math8]

[00008]
$$c^j = \{h_r^j, \ {}_r^j, \hat{h}^j, \hat{\ }^j, X^j\} \quad (10)$$

[0321] The training data c.sup.j is a combination of input data h{circumflex over ( )}.sup.j, κ{circumflex over ( )}.sup.j, and X.sup.j, and the shape h.sub.r.sup.j of the earth and the characteristic κ.sub.r.sup.j of the earth at the time of completion of the path X.sup.j as the ground truth. The input data h{circumflex over ( )}.sup.j, κ{circumflex over ( )}.sup.j, and X.sup.j correspond to the input data of the function g in equations (7) and (8).
[0322] As described above, the training data set D may be generated from the log acquired by the log acquisition part **2001**, the log acquired by the simulator part **2002**, or both logs.
[0323] In the simulator part **2002**, for example, as described above, a particle simulation such as DEM is adopted, and the height h.sub.r.sup.j of the earth is acquired by ray tracing of the shape sensor such as LIDAR which is virtually arranged with respect to the position of the particle.
[0324] As described above, the training data set D may include a base training data set generated from logs acquired by the simulator part **2002** and a fine-tuning training data set generated from logs acquired by the log acquisition part **2001**. In this case, the number of training data included in the fine-tuning training data set may be relatively small.
[0325] As shown in the following equation (11), the function g has a parameter W, and the machine learning part **2005**A performs machine learning in such a manner that the parameter W is optimized by the training data set D.

[Math9]

[00009]
$$(h^j, \ ^j) = g(\hat{h}^j, \hat{\ }^j, X^j; w) \quad (11)$$

[0326] For example, the function g corresponding to the trained model LM**1** is generated by optimizing the parameter W so that the loss function E(W) in the following equation (12) is minimized.

[Math10]

[00010]
$$E(w) = -\ \text{.Math.}_j \log N(h_r^j \ \text{.Math.}\ h^j, \ ^j; W) \quad (12)$$

[0327] Thus, the information processing device **200** generates the training data set D including the training data c.sup.j, and can generate the function g corresponding to the trained model LM**1** by machine learning based on the training data set D.
[Action]
[0328] Next, the action of the working machine, the information processing device, and the

program according to the present embodiment will be described.

[0329] In the present embodiment, the working machine includes an operation planning part. The working machine is, for example, the above-described shovel **100**. The operation planning part is, for example, the above-described operation planning part **302**C. Specifically, the operation planning part determines the future operation of the working machine from among a plurality of according operations to the performance status of the operation of the working machine.

[0330] In the present embodiment, the information processing device may include an operation planning part. The information processing device may be, for example, the controller **30**, the information processing device **200**, or the remote control support device **400** described above.

[0331] In the present embodiment, the program may cause the information processing device to perform an operation planning step in order to achieve the function of the operation planning part. Specifically, in the operation planning step, the future operation of the working machine is determined from among a plurality of operations according to the performance status of the operation of the working machine. The operation planning step is, for example, the step S**206** described above.

[0332] Thus, the working machine or the like can determine the future operation of the working machine from among a plurality of operations in consideration of the state of the target material which changes according to the performance status of the operation of the working machine. Therefore, the working machine can achieve a more appropriate operation.

[0333] In the present embodiment, the working machine and the information processing device may include a prediction part. The prediction part is, for example, the target material state prediction part **302**B described above. Specifically, the prediction part may predict a future state of the target material according to the performance status of the operation of the working machine. The operation planning part may determine the future operation of the working machine from among the plurality of operations, based on the prediction result of the future state of the target material by the prediction part.

[0334] In the present embodiment, the program may cause the information processing device to perform the prediction step in order to achieve the function of the prediction part. Specifically, the prediction step may predict the future state of the target material according to the performance state of the operation of the working machine. The operation planning step may decide the future operation of the working machine from among a plurality of operations based on the prediction result of the future state of the target material by the prediction part.

[0335] Thus, the working machine or the like can determine the future operation of the working machine from among a plurality of operations in accordance with the future state of the target material predicted according to the performance status of the operation of the working machine.

[0336] In the present embodiment, the prediction part may predict the future state of the target material, based on the path of the work part by the operation of the working machine.

[0337] Thus, the working machine or the like can predict the future state of the target material considering the change of the state of the target material which changes in accordance with the path of the work part.

[0338] In the present embodiment, the prediction part may predict a state of the target material after completion of a current operation of the working machine or a next operation after the current operation, based on the path of the work part by the current operation of the working machine or a next scheduled operation. Then, the operation planning part determines a next operation after the current operation of the working machine or a further next operation after the next scheduled operation from among the plurality of operations, based on the prediction result of the future state of the target material by the prediction part.

[0339] Thus, the working machine or the like can determine the next operation according to the prediction result of the work state after the completion of the operation under performance of the working machine. Furthermore, the working machine or the like can determine the next operation

after the next operation scheduled to be performed according to the prediction result of the work state after the completion of the next operation.
[0340] In addition, in the present embodiment, the working machine or the like may include a generation part and a control part. The generation part is, for example, the target path generation part **302**D described above. The control part is, for example, the operation control part **302**E described above. Specifically, the generation part may generate the path of the work part by the operation of the working machine, based on the state of the target material. The control part may control the operation of the working machine so that the work part moves along the path generated by the generation part. The prediction part may predict the future state of the target material, based on the current state of the target material and the path of the work part generated by the generation part.
[0341] In the present embodiment, the program may cause the information processing device to perform the generation step and the control step in order to achieve the functions of the generation part and the control part. The generation step is, for example, the step S**208** described above. The control step is, for example, the step S**404** described above. Specifically, in the generation step, the path of the work part by the operation of the working machine may be generated based on the state of the target material. In the control step, the operation of the working machine may be controlled so that the work part moves along the path generated in the generation step. In the prediction step, the future state of the target material may be predicted based on the current state of the target material and the path of the work part generated in the generation step.
[0342] Thus, the working machine or the like can predict the future state of the target material based on the path of the work part generated for automatic operation of the working machine.
[0343] In the present embodiment, the prediction part may predict the state of the target material after the elapse of a predetermined time. The generation part may generate the path of the work part after the elapse of the predetermined time, based on the prediction result of the state of the target material after the elapse of the predetermined time by the prediction part.
[0344] Thus, the working machine or the like can generate the path of the work part after the elapse of the predetermined time considering, for example, the delay time from the instruction to control the operation of the working machine to the actual control of the operation.
[0345] Further, in the present embodiment, the generation part may generate the path of the work part, based on measurement data of the state of the target material, by using an objective function and a constraint function defined for each of the plurality of operations.
[0346] Thus, the working machine or the like can generate the path of the work part.
[0347] Further, in the present embodiment, the path of the work part may be expressed by a predetermined number, which is two or greater, of parameters for each of the plurality of operations. Then, the generation part may generate the path of the work part by determining a predetermined number of parameters, based on the state of the target material.
[0348] Thus, the working machine or the like can generate the path of the work part.
[0349] In the present embodiment, the state of the target material may include at least one of the shape and the characteristic of the earth on the surface of the target material.
[0350] Thus, the working machine or the like can predict the future state of the target material in accordance with the shape and the characteristic of the earth of the target material.
[0351] In the present embodiment, the working machine or the like may include a notification part for notifying the operator of the operation determined by the operation planning part.
[0352] In the present embodiment, the program may cause the support device to perform the operation planning step and the notification step. The support device is, for example, the support device **150** and the remote control support device **400**. Specifically, in the operation planning step, the future operation of the working machine may be decided from among a plurality of operations according to the performance status of the operation of the working machine. In the notification step, the operation decided in the operation planning step may be notified to the operator of the

working machine.
[0353] Thus, the working machine or the like can notify the operator of a more appropriate type of operation of the working machine to the operator who operates the working machine. Therefore, the operator can operate the working machine more appropriately. Moreover, the working machine or the like can notify the operator in advance of a more appropriate type of operation of the working machine to the operator who operates the working machine. Therefore, the operator can start the next operation immediately after the completion of a certain operation, and as a result, the work efficiency of the working machine can be improved.
[0354] Further, the present disclosure is not limited to these embodiments, but various variations and modifications may be made without departing from the scope of the present disclosure.

Claims

1. An information processing device comprising an operation planning part configured to determine a future operation of a working machine from among a plurality of operations according to a performance status of an operation of the working machine.
2. A working machine comprising the information processing device according to claim 1, and an operation planning part configured to determine a future operation of the working machine from among a plurality of operations according to a performance status of an operation of the working machine.
3. The working machine according to claim 2, further comprising a prediction part configured to predict a future state of a target material according to the performance status of the operation of the working machine, the operation part wherein planning is configured to determine the future operation of the working machine from among the plurality of operations, based on a prediction result of the future state of the target material by the prediction part.
4. The working machine according to claim 3, wherein the prediction part is configured to predict the future state of the target material, based on a path of a work part by an operation of the working machine.
5. The working machine according to claim 4, wherein: the prediction part is configured to predict a state of the target material after completion of a current operation of the working machine or a next operation after the current operation, based on the path of the work part by the current operation of the working machine or a next scheduled operation; and the operation planning part is configured to determine the next operation after the current operation of the working machine or a further next operation after the next scheduled operation from among the plurality of operations, based on the prediction result of the future state of the target material by the prediction part.
6. The working machine according to claim 4, further comprising: a generation part configured to generate a path of the work part by an operation of the working machine, based on a state of the target material; and a control part configured to control an operation of the working machine so that the work part moves along the path generated by the generation part, wherein the prediction part is configured to predict the future state of the target material, based on a current state of the target material and the path of the work part generated by the generation part.
7. The working machine according to claim 6, wherein: the prediction part is configured to predict the state of the target material after a elapse of a predetermined time; and the generation part is configured to generate the path of the work part after the elapse of the predetermined time, based on the prediction result of the state of the target material after the elapse of the predetermined time by the prediction part.
8. The working machine according to claim 6, wherein the generation part is configured to generate the path of the work part, based on measurement data of the state of the target material, by using an objective function and a constraint function defined for each of the plurality of operations.
9. The working machine according to claim 6, wherein: the path of the work part is expressed by a

predetermined number of parameters for each of the plurality of operations, the predetermined number being two or greater; and the generation part is configured to generate the path of the work part by determining the predetermined number of parameters, based on the state of the target material.

10. The working machine according to claim 3, wherein the state of the target material includes at least one of a shape and a characteristic of earth on a surface of the target material.

11. The working machine according to claim 2, further comprising a notification part configured to notify an operator of the future operation determined by the operation planning part.

12. A non-transitory computer-readable recording medium having a program embodied therein for causing an information processing device to determine a future operation of a working machine from among a plurality of operations according to a performance status of an operation of the working machine.

13. A non-transitory computer-readable recording medium having a program embodied therein for causing a support device to perform: determining a future operation of a working machine from among a plurality of operations according to a performance status of an operation of the working machine; and notifying an operator of the determined future operation.

14. The working machine according to claim 2, wherein the plurality of operations are of different types, and each type is predetermined.

15. The working machine according to claim 3, further comprising a storage part configured to store a trained model for outputting one operation from among the plurality of operations, when the state of the target material is an input, wherein the operation planning part is configured to determine the future operation of the working machine from among the plurality of operations by using the trained model, based on the prediction result of the future state of the target material by the prediction part.

16. The working machine according to claim 3, wherein: a candidate operation capable of transitioning as a next operation within the plurality of operations, starting from a target operation by each of the plurality of operations, and a transition condition for transitioning to the candidate operation are predetermined; and the operation planning part is configured to determine the future operation of the working machine from among the plurality of operations by using the transition condition, based on the prediction result of the future state of the target material by the prediction part.

17. The working machine according to claim 3, wherein: the prediction part is configured to predict a state of the target material at a predetermined time in future, based on the operation status of the working machine; and the operation planning part is configured to determine a future operation starting from the predetermined time, based on the prediction result of the state of the target material at the predetermined time in the future by the prediction part.

18. The working machine according to claim 2, wherein the operation planning part is configured to determine an undetermined future operation of the working machine from among a plurality of operations according to the performance status of the operations of the working machine.

19. The working machine according to claim 3, further comprising a storage part configured to store a trained model for outputting one operation from among the plurality of operations, when information concerning the performance status of the operation of the working machine is an input, wherein the prediction part is configured to predict the future state of the target material, based on the information concerning the performance status of the operation of the working machine, using the trained model.